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Electric circuit breaker

Abstract

An electric circuit breaker that includes a housing, a cut portion that is disposed in the housing and constitutes a part of an electric circuit, a first power source that is disposed on a side of a first end portion of the housing, and a moving body that moves in the housing between the first end portion and a second end portion opposite to the first end portion, the electric circuit breaker including a fuse function portion that includes a fusion portion and an arc-extinguishing material, wherein the moving body is configured to cut a cut piece positioned between base pieces on both sides of the cut portion at a part of the moving body while moving from the first end portion toward the second end portion by the first power source.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This is a 35 U.S.C. § 371 application of, and claims priority to, International Application No. PCT/JP2022/018346, filed on Apr. 21, 2022, which was published as WO2022/264686A1 on Dec. 22, 2022, and which claims priority to JP Patent Application No. 2021-100645, which was filed on Jun. 17, 2021, the teachings of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to an electric circuit breaker that can be mainly used for an electric circuit of an automobile or the like.

BACKGROUND ART

(3) Conventionally, an electric circuit breaker has been used to protect an electric circuit mounted on an automobile or the like and various electric components connected to the electric circuit. Specifically, in a case where an abnormality occurs in the electric circuit, the electric circuit breaker disconnects a part of the electric circuit to physically break the electric circuit.

(4) In addition, a voltage and a current applied to the electric circuit tend to increase due to recent improvement in performance of automobiles and the like, and it has been required to extinguish an arc generated immediately after the electric circuit is broken by the electric circuit breaker more effectively, quickly, and safely. Therefore, the electric circuit breaker according to Patent Literature 1 is an electric circuit breaker including a fuse, a housing, a cut portion that is disposed in the housing and constitutes a part of an electric circuit, a power source that is disposed on a first end portion side of the housing, and a moving body that moves in the housing between a first end portion and a second end portion opposite to the first end portion, in which the moving body is moved by the power source from the first end portion toward the second end portion, and a part of the moving body cuts the cut portion to break the electric circuit. A current (fault current) flowing through the electric circuit when the electric circuit is broken is induced in the fuse, and the arc

generated by the induced current is effectively, quickly, and safely extinguished in the fuse.

(5) Furthermore, the current to be broken in the electric circuit is assumed to be not only a relatively high current but also in a wide range up to a relatively low current. Therefore, in the electric circuit breaker of Patent Literature 1, in a case where the current (fault current) induced when the electric circuit is broken is relatively low, depending on the fusing characteristics of the fuse, the time until the fuse breaks the current may be long or the current may not be broken.

CITATION LIST

Patent Literature

(6) Patent Literature 1: Japanese Patent Application No. 2020-208249

SUMMARY OF INVENTION

Technical Problems

(7) Therefore, in view of the above problems, the present invention provides an electric circuit breaker that quickly breaks current in a wide current range up to a relatively low current as well as a relatively high current.

Solution to Problems

(8) An electric circuit breaker of the present invention is an electric circuit breaker that includes a housing, a cut portion that is disposed in the housing and constitutes a part of an electric circuit, a first power source that is disposed on a first end portion side of the housing, and a moving body that moves in the housing between the first end portion and a second end portion opposite to the first end portion, the electric circuit breaker including a fuse function portion that includes a fusion portion and an arc-extinguishing material, wherein the moving body is configured to cut a cut piece positioned between base pieces on both sides of the cut portion at a part of the moving body while moving from the first end portion toward the second end portion by the first power source, in a case where a current to be broken is low, the fuse function portion and the cut portion are not connected, the moving body is moved toward the second end portion by the first power source to cut the cut piece positioned between the base pieces on both sides of the cut portion to break a state where the base pieces on both sides of the cut portion are energized, and in a case where the current to be broken is high, the fuse function portion and the cut portion are connected to each other, the moving body is moved toward the second end portion by the first power source to cut the cut piece positioned between the base pieces on both sides of the cut portion to break the state where the base pieces on both sides of the cut portion are energized.

(9) The electric circuit breaker according to the present invention further includes paired electrodes individually connected to terminals on both sides of the fuse function portion, wherein in a case where a current to be broken is low, the moving body moves toward the second end portion to cut a cut piece positioned between base pieces on both sides of the cut portion so as to break a state where the base pieces on both sides of the cut portion are energized, and a regulating unit operated by a second power source regulates movement of the moving body so as not to connect a part of the cut portion and the electrode in order to make the cut portion and the fuse function portion unconnected, in a case where the current to be broken is high, the moving body moves toward the second end portion, and in a state where the base pieces on both sides of the cut portion are energized via the cut piece, a part of the cut portion and the electrode come into contact with each other to connect the cut portion and the fuse function portion, and thereafter, the state where the base pieces on both sides of the cut portion are energized via the cut piece is broken along with the movement of the moving body.

(10) Furthermore, in the electric circuit breaker according to the present invention, the moving body includes the electrode, a state where base pieces on both sides of the cut portion are energized via the cut piece is a state where the base piece and the cut piece physically cut and separated from the base piece are energized by arc discharge, and the energized state is broken by an insulator being interposed between the base piece and the cut piece along with movement of the moving body.

(11) Moreover, in the electric circuit breaker according to the present invention, the housing includes the electrode, a state where base pieces on both sides of the cut portion are energized via the cut piece is a state where the base piece and the cut piece physically cut and separated from the base piece are energized by a conductor included in the moving body, and in the energized state, the base piece of the cut portion and the electrode are connected via the conductor of the moving body, and the cut portion and the fuse function portion are connected.

(12) The electric circuit breaker according to the present invention further includes a circuit connected to the cut portion via the fuse function portion, wherein in a case where a current to be broken is low, the circuit is broken by a breaker moved by a second power source to be in a state where the fuse function portion and the cut portion are not connected, and thereafter, the moving body is moved toward the second end portion by a first power source to cut a cut piece positioned between base pieces on both sides of the cut portion so as to break a state where the base pieces on both sides of the cut portion are energized, and in a case where the current to be broken is high, in a state where the circuit is not broken and the fuse function portion and the cut portion remain connected to each other, the moving body is moved toward the second end portion by the first power source to cut the cut piece positioned between the base pieces on both sides of the cut portion so as to break the state where the base pieces on both sides of the cut portion are energized.

(13) In addition, in the electric circuit breaker according to the present invention, a fuse element of the fuse function portion constitutes a part of the circuit, the fuse element is surrounded by an arc-extinguishing material, and in a case where a current to be broken is low, the fuse element that is a part of the circuit is broken by a breaker moved by the second power source.

(14) According to each of the above features, in a case where an overcurrent belonging to a relatively low current range flows through the electric circuit, the fuse function portion and the cut portion are not connected, and the moving body moves toward the second end portion by the first power source to cut the cut piece positioned between the base pieces on both sides of the cut portion, so that the state where the base pieces on both sides of the cut portion are energized is broken and the overcurrent is prevented from flowing through the electric circuit. Therefore, it is possible to solve the problem that, as in a conventional case, the current belonging to the relatively low current range cannot be broken because the fusion portion of the fuse function portion is not fused or the overcurrent flowing through the electric circuit cannot be broken immediately because it takes a relatively long time to break the current. On the other hand, in a case where an overcurrent belonging to a relatively high current range flows through the electric circuit, the fuse function portion and the cut portion are connected, the moving body moves toward the second end portion by the first power source to cut the cut piece positioned between the base pieces on both sides of the cut portion, so that the state where the base pieces on both sides of the cut portion are energized is broken and the overcurrent is safely prevented from flowing through the electric circuit. As described above, the electric circuit breaker of the present invention quickly breaks current in a wide current range up to a relatively low current as well as a relatively high current.

Advantageous Effects of Invention

(15) As described above, the electric circuit breaker of the present invention quickly breaks current in a wide current range up to a relatively low current as well as a relatively high current.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1(a) is an overall perspective view of a lower housing constituting a housing of an electric circuit breaker according to a first embodiment of the present invention, FIG. 1(b) is a plan view of the lower housing, and FIG. 1(c) is a cross-sectional view taken along line A-A.

(2) FIG. 2(a) is an overall perspective view of an upper housing constituting the housing of the

electric circuit breaker according to the first embodiment of the present invention, FIG. 2(b) is a plan view of the upper housing, and FIG. 2(c) is a cross-sectional view of the upper housing taken along line B-B.

(3) FIG. 3(a) is an exploded perspective view of a moving body of the electric circuit breaker according to the first embodiment of the present invention, FIG. 3(b) is a perspective view of the moving body, and FIG. 3(c) is a cross-sectional view taken along line C-C.

(4) FIG. 4(a) is a perspective view of a cut portion of the electric circuit breaker according to the first embodiment of the present invention, and FIG. 4(b) is a cross-sectional view taken along line D-D.

(5) FIG. 5 is an exploded perspective view of the electric circuit breaker according to the first embodiment of the present invention.

(6) FIG. 6 is a cross-sectional view taken along line E-E in a state where the electric circuit breaker according to the first embodiment of the present invention is assembled.

(7) FIG. 7 is a cross-sectional view taken along line F-F in a state where the electric circuit breaker illustrated in FIG. 5 is assembled.

(8) FIG. 8 is a cross-sectional view illustrating a state where the moving body has moved from the state illustrated in FIG. 6.

(9) FIG. 9 is a cross-sectional view illustrating a state where the moving body has moved from the state illustrated in FIG. 7.

(10) FIG. 10 is a cross-sectional view illustrating a state where the moving body has moved from the state illustrated in FIG. 6.

(11) FIG. 11 is a cross-sectional view illustrating a state where the moving body has moved from the state illustrated in FIG. 6.

(12) FIG. 12 is a cross-sectional view illustrating a state where the moving body has moved from the state illustrated in FIG. 6.

(13) FIG. 13(a) is an overall perspective view of a lower housing of an electric circuit breaker according to a second embodiment of the present invention, FIG. 13(b) is a plan view of the lower housing, and FIG. 13(c) is a cross-sectional view taken along line G-G.

(14) FIG. 14 is an exploded perspective view of the electric circuit breaker according to the second embodiment of the present invention.

(15) FIG. 15 is a cross-sectional view taken along line H-H in a state where the electric circuit breaker illustrated in FIG. 14 is assembled.

(16) FIG. 16 is a cross-sectional view illustrating a state where a moving body has moved from the state illustrated in FIG. 15.

(17) FIG. 17(a) is an overall perspective view of a lower housing of an electric circuit breaker according to a third embodiment of the present invention, FIG. 17(b) is a plan view of the lower housing, and FIG. 17(c) is a cross-sectional view taken along line I-I.

(18) FIG. 18 is an exploded perspective view of the electric circuit breaker according to the third embodiment of the present invention.

(19) FIG. 19 is a cross-sectional view taken along line J-J in a state where the electric circuit breaker illustrated in FIG. 18 is assembled.

(20) FIG. 20 is a cross-sectional view taken along line K-K in a state where the electric circuit breaker illustrated in FIG. 18 is assembled.

(21) FIG. 21 is a cross-sectional view illustrating a state where a moving body has moved from the state illustrated in FIG. 19.

(22) FIG. 22 is a cross-sectional view illustrating a state where the moving body has moved from the state illustrated in FIG. 20.

(23) FIG. 23 is a cross-sectional view illustrating a state where the moving body has moved from the state illustrated in FIG. 19.

(24) FIG. 24 is a cross-sectional view illustrating a state where the moving body has moved from

the state illustrated in FIG. 19.

(25) FIG. 25 is a cross-sectional view illustrating a state where the moving body has moved from the state illustrated in FIG. 19.

(26) FIG. 26 is an overall perspective view of an electric circuit breaker according to a fourth embodiment of the present invention in an exploded manner.

(27) FIG. 27(a) is a cross-sectional view of FIG. 26 taken along line S-S, and FIG. 27(b) is a cross-sectional view of FIG. 26 taken along line L-L.

(28) FIG. 28(a) is a cross-sectional view illustrating a state where a breaker has moved from the state illustrated in FIG. 27(b), and FIG. 28(b) is a cross-sectional view illustrating a state where a moving body has moved from the state illustrated in FIG. 28(a).

(29) FIG. 29 is a cross-sectional view illustrating a state where the moving body has moved from the state illustrated in FIG. 27(b).

(30) FIG. 30 is an overall perspective view of an electric circuit breaker according to a fifth embodiment of the present invention in an exploded manner.

(31) FIG. 31(a) is a cross-sectional view of FIG. 30 taken along line N-N, and FIG. 31(b) is a cross-sectional view of FIG. 30 taken along line M-M.

(32) FIG. 32(a) is a cross-sectional view illustrating a state where a breaker has moved from the state illustrated in FIG. 31(b), and FIG. 32(b) is a cross-sectional view illustrating a state where a moving body has moved from the state illustrated in FIG. 32(a).

(33) FIG. 33 is a cross-sectional view illustrating a state where the moving body has moved from the state illustrated in FIG. 31(b).

REFERENCE SIGNS LIST

(34) **300** housing **320** first end portion **330** second end portion **400** cut portion **420** cut piece **430** base piece **500** moving body **600** electric circuit breaker **700** fuse function portion **730** arc-extinguishing material **740** fusion portion P power source

DESCRIPTION OF EMBODIMENTS

(35) Hereinafter, each embodiment of the present invention will be described with reference to the drawings. Note that, the shape, material, and the like of each member of an electric circuit breaker according to embodiments described below are merely examples, and are not limited thereto.

First Embodiment

(36) First, FIG. 1 illustrates a lower housing **100** constituting a housing **300** of an electric circuit breaker according to a first embodiment of the present invention. Note that FIG. 1(a) is an overall perspective view of the lower housing **100**, FIG. 1(b) is a plan view of the lower housing **100**, and FIG. 1(c) is a cross-sectional view taken along line A-A.

(37) As illustrated in FIG. 1, the lower housing **100** is a substantially quadrangular prism formed of an insulator such as a synthetic resin, and includes a hollow lower housing portion **110** therein. The lower housing portion **110** extends from an upper surface **120** toward a lower surface **130** of the lower housing **100**, and is configured to house a moving body **500** to be described later. In addition, an inner surface **111** of the lower housing portion **110** is a smooth surface so that the moving body **500** can slide therein in a vertical direction. Furthermore, placement portions **113** recessed based on the shape of a base piece **430** are provided in a part of the upper surface **120** so that the base piece **430** of a cut portion **400** to be described later can be placed. The placement portions **113** are arranged on both sides of the lower housing portion **110** so as to face each other, and support the cut portion **400** extending linearly on both sides. In addition, the placement portion **113** includes claws **114**, and can be engaged with a part of the base piece **430** of the cut portion **400** placed to fix the cut portion **400** without any deviation. Moreover, coupling bores B1 are formed at four corners of the upper surface **120** of the lower housing **100**, and these coupling bores B1 are arranged so as to vertically match coupling bores B2 of an upper housing **200** to be described later.

(38) Next, FIG. 2 illustrates the upper housing **200** constituting the housing **300** according to the first embodiment of the present invention. FIG. 2(a) is an overall perspective view of the upper

housing **200**, FIG. 2(b) is a plan view of the upper housing **200**, and FIG. 2(c) is a cross-sectional view of the upper housing **200** taken along line B-B.

(39) As illustrated in FIG. 2, the upper housing **200** is a substantially quadrangular prism formed of an insulator such as a synthetic resin, and forms a pair with the lower housing **100** illustrated in FIG. 1 to constitute the housing **300**. The upper housing includes an upper housing portion **210** therein, the upper housing portion **210** extends from a lower surface **230** toward an upper surface **220** of the upper housing **200**, and is configured to house the moving body **500** to be described later. In addition, an inner surface **211** of the upper housing portion **210** is a smooth surface so that the moving body **500** can slide therein in the vertical direction. As described later, the upper housing portion **210** is arranged vertically with the lower housing portion **110** of the lower housing **100** to constitute a housing portion **310** extending linearly, and the moving body **500** can move vertically in the housing portion **310**.

(40) Furthermore, insertion portions **213** recessed based on the shape of the base piece **430** are provided in a part of the lower surface **230** so that the base piece **430** of the cut portion **400** to be described later can be inserted. The insertion portions **213** are arranged on both sides of the upper housing portion **210** so as to face each other, and are arranged at positions corresponding to the placement portions **113** of the lower housing **100**. Therefore, the insertion portion **213** is fitted from above to the base piece **430** of the cut portion **400** placed on the placement portion **113** of the lower housing **100**.

(41) Moreover, a power source storage portion **221** in which a first power source P is housed is formed in a part of the side of the upper surface **220** of the upper housing **200**. The power source storage portion **221** communicates with the upper end side of the upper housing portion **210**. As described in detail later, power such as air pressure generated from the first power source P housed in the power source storage portion **221** is transmitted to the moving body **500** in the upper housing portion **210** to move the moving body **500**. The lower housing **100** and the upper housing **200** are substantially quadrangular prisms formed of a synthetic resin, but are not limited thereto, and may have any shape formed of other materials as long as they have high insulating properties and strength enough to withstand use.

(42) Furthermore, a regulating unit **800** formed of an insulator such as a synthetic resin is attached to the upper housing **200**. The regulating unit **800** includes a housing portion **820** in which a regulating body **810** is slidably housed, and a second power source **830** for moving the regulating body **810**. The regulating body **810** includes a terminal portion **812** and a rod-shaped distal end portion **811** extending in an elongated shape from the terminal portion **812**. In addition, the upper housing **200** includes a through-hole **250** that allows the upper housing portion **210** in the upper housing and the housing portion **820** of the regulating unit **800** to communicate with each other, and the through-hole **250** is formed in a manner that the distal end portion **811** of the regulating body **810** of the regulating unit **800** can be inserted therethrough. As described in detail later, power such as air pressure generated from the second power source **830** housed in the housing portion **820** is transmitted to the regulating body **810** in the housing portion **820** to move the regulating body **810** toward the upper housing portion **210** of the upper housing **200**. Then, the distal end portion **811** of the regulating body **810**, which has moved, passes through the through-hole **250** and moves into the upper housing portion **210** of the upper housing **200**. Note that the regulating unit **800** is attached to the upper housing **200**, but is not limited thereto, and can be attached to any place as long as it is a part of the housing **300**.

(43) Next, the moving body **500** according to the first embodiment of the present invention is illustrated in FIG. 3. Note that FIG. 3(a) is an exploded perspective view of the moving body **500**, FIG. 3(b) is a perspective view of the moving body **500**, and FIG. 3(c) is a cross-sectional view taken along line C-C.

(44) As illustrated in FIG. 3, the moving body **500** is formed of an insulator such as a synthetic resin, and includes a substantially cylindrical body **510** on the upper end side, a flat quadrangular

sliding portion **520** in the center, and a projection **530** projecting downward on the lower end side. A recessed portion **511** is provided at the upper end of the body **510**, and the recessed portion **511** is a portion facing the first power source P. The sliding portion **520** has a shape corresponding to the inner surface shape of the housing portion **310**, and the sliding portion **520** slides on the inner surface of the housing portion **310**, so that the moving body **500** can smoothly slide while maintaining a posture along the inner side of the housing portion **310**. In addition, a part of the sliding portion **520** has a stepped shape, and includes an abutting portion **521** capable of abutting on a part of the regulating body **810** of the regulating unit **800**. Note that a groove **514** is formed on the outer periphery of a part of the body **510**, and an O-ring **515** (elastically deformable synthetic resin ring) is fitted into the groove **514**. Therefore, as described later, the air pressure due to the explosion of the first power source P is prevented from leaking from the space formed by the recessed portion **511**.

(45) Furthermore, two plate-like electrodes **540** and **550** are fixed to both sides of the projection **530**. The paired electrodes (**540**, **550**) are connected to terminals of a fuse function portion to be described later, and are formed of a metal conductor such as copper so as to be electrically connected to a part of the cut portion **400**. Since the electrode **540** and the electrode **550** are fixed to both sides with the projection **530** formed of an insulator interposed therebetween, the electrode **540** and the electrode **550** are not electrically connected to each other and are in an independent state. In addition, the moving body **500** includes a plate-like insulator **560** formed of a synthetic resin, ceramics, or the like on the distal end side of the electrode **540** and the electrode **550**.

(46) Note that the moving body **500** is formed of a synthetic resin, but is not limited thereto, and may have any shape formed of other materials as long as they have high insulating properties and strength enough to withstand use. Furthermore, the paired electrodes **540** and **550** are formed in a plate shape, but are not limited thereto, and may have any shape as long as they can be electrically connected to a part of the cut portion **400**.

(47) Next, FIG. **4** illustrates the cut portion **400** constituting a part of an electric circuit to be broken by an electric circuit breaker **600** according to the first embodiment of the present invention. Note that FIG. **4(a)** is a perspective view of the cut portion **400**, and FIG. **4(b)** is a cross-sectional view taken along line D-D.

(48) The cut portion **400** is entirely a metal conductor such as copper in order to be electrically connected to the electric circuit, and includes the base piece **430** for connecting to the electric circuit at both ends and a cut piece **420** positioned between the base pieces **430**. A connection hole **410** used for connection with the electric circuit is formed at an end portion of the base piece **430**. In addition, a linear cut **424** is formed in a back surface **421** of the boundary portion between the cut piece **420** and the base piece **430** so as to traverse in the width direction of the cut portion **400** in order to facilitate cutting of the cut piece **420** from the base piece **430**. Note that the cut portion **400** is not limited to the shape illustrated in FIG. **4**, and may have any shape as long as it includes the base piece **430** for electrically connecting to the electric circuit and the cut piece **420** positioned between the base pieces **430**. In addition, although the cross-sectional area of a part of the cut piece **420** is minimized by the cut **424** to facilitate cutting, the shape and position of the cut **424** can be appropriately changed depending on the configuration of the moving body **500** to facilitate cutting by the moving body **500**.

(49) Next, a method of assembling the electric circuit breaker **600** of the present invention will be described with reference to FIG. **5**. Note that FIG. **5** is an exploded perspective view of the electric circuit breaker **600**.

(50) When the electric circuit breaker **600** is assembled, first, an abutment base **112** formed of an insulator is fixed to the bottom of the lower housing portion **110** of the lower housing **100**. Next, the base piece **430** of the cut portion **400** is placed on the placement portion **113** of the lower housing **100**, and the cut portion **400** is disposed in a manner that the cut piece **420** traverses the lower housing portion **110** of the lower housing **100**.

(51) Next, the upper housing **200** is fitted from above the lower housing **100** in a manner that the side of the body **510** of the moving body **500** is inserted into the upper housing portion **210** of the upper housing **200**. Then, the insertion portion **213** of the upper housing **200** is fitted to the base piece **430** of the cut portion **400**. By coupling and fixing the coupling bore **B1** and the coupling bore **B2** aligned vertically using a coupling tool or the like, the housing **300** including the lower housing **100** and the upper housing **200** is assembled in a state where the cut portion **400** and the moving body **500** are housed therein.

(52) Furthermore, the first power source **P** is attached to the power source storage portion **221** of the upper housing **200**, and a part of the first power source **P** is housed in the recessed portion **511** of the moving body **500**. In addition, when it is detected that an abnormal current flows through the electric circuit, an abnormality signal is input from an external device to the first power source **P**. Then, for example, the gunpowder in the first power source **P** is exploded, and the moving body **500** is instantaneously pushed out and moved in the housing portion **310** by the air pressure due to the explosion. Note that the first power source **P** is not limited to a power source using gunpowder as long as it generates power for moving the moving body **500**, and other known power sources may be used.

(53) The regulating unit **800** is attached to the upper housing **200**. A part of the regulating body **810** is configured to be movable into the upper housing portion **210** of the upper housing **200** by the second power source **830**. In addition, when it is detected that an abnormal current flows through the electric circuit and an abnormality signal is input from an external device to the second power source **830**, for example, the gunpowder in the second power source **830** is exploded, and the regulating body **810** is instantaneously pushed out and moved in the housing portion **820** by the air pressure due to the explosion. Note that the second power source **830** is not limited to a power source using gunpowder as long as it generates power for moving the regulating body **810**, and other known power sources may be used.

(54) The electric circuit breaker **600** also includes a fuse function portion **700**. The fuse function portion **700** includes a fuse element **720** formed of a conductive metal such as copper or an alloy thereof in a hollow and insulating casing **710**, and the periphery of the fuse element **720** inside the casing **710** is filled with an arc-extinguishing material **730**. Terminals **750** on both sides of the fuse element **720** are electrically connected to the paired electrodes **540** and **550** by connection members **760** such as electric wires. In addition, the fuse element **720** includes a fusion portion **740** between the terminals **750**, and the fusion portion **740** is a portion in which the width of the fuse element **720** is locally narrowed, and is configured to generate heat and fuse to break the current when the current to be broken by the electric circuit breaker flows.

(55) The arc-extinguishing material **730** is a granular arc-extinguishing material made of silica sand or the like, or a gaseous arc-extinguishing material made of nitrogen gas or the like, and is configured to extinguish the arc generated between the terminals **750** after the fusion of the fusion portion **740**. As the fuse function portion **700**, an existing fuse that is conventionally known and in which an arc-extinguishing material and a fuse element are enclosed in a casing can be used, and a fuse having arc-extinguishing performance based on a current or a voltage to be broken by the electric circuit breaker can be appropriately adopted. Note that the fuse function portion **700** can be attached to any place in the housing **300**. By attaching the fuse function portion **700** to the housing **300**, the fuse function portion **700** is less likely to be affected by an impact due to the movement of the moving body **500** and is less likely to be damaged.

(56) Next, an internal structure of the electric circuit breaker **600** according to the first embodiment of the present invention will be described with reference to FIGS. **6** and **7**. Note that FIG. **6** is a cross-sectional view taken along line E-E and FIG. **7** is a cross-sectional view taken along line F-F in a state where the electric circuit breaker **600** illustrated in FIG. **5** is assembled.

(57) As illustrated in FIG. **6**, the moving body **500** is housed in the housing portion **310** including the lower housing portion **110** and the upper housing portion **210** that are linearly arranged. The

housing portion **310** extends from a first end portion **320** of the housing **300** to a second end portion **330** opposite to the first end portion **320**. Since the moving body **500** is disposed on the side of the first end portion **320** in which the first power source P is disposed, the side of the second end portion **330** of the housing portion **310** is hollow. Therefore, as described later, the moving body **500** can move toward the second end portion **330** while cutting the cut piece **420**. In addition, since the recessed portion **511** on the upper end side of the moving body **500** is adjacent to the first power source P, the air pressure due to the explosion of the gunpowder in the first power source P is transmitted to the upper end side of the moving body **500** as described later.

(58) Note that, as illustrated in FIG. 6, the assembled and completed electric circuit breaker **600** is attached in an electric circuit to be protected and used. Specifically, the base piece **430** of the cut portion **400** is connected to a part of the electric circuit, and the cut portion **400** constitutes a part of the electric circuit. The insulator **560** extends along the cut piece **420** and is disposed away from the cut piece **420**. In a normal state (that is, when no abnormal current flows), since the base piece **430** and the cut piece **420** of the cut portion **400** are not cut and are physically and electrically connected, a current I1 flows through the electric circuit via the base piece **430** and the cut piece **420** of the cut portion **400**. In addition, as illustrated in FIG. 7, the distal end portion **811** of the regulating body **810** of the regulating unit **800** is inserted into the through-hole **250** of the housing **300** but does not project to the housing portion **310**. Therefore, in the normal state, a part of the regulating body **810** of the regulating unit **800** does not abut on the abutting portion **521** of the moving body **500**, and the regulating unit **800** does not regulate the movement of the moving body **500**.

(59) Furthermore, as illustrated in FIG. 7, a device S that detects an abnormal current in the electric circuit is connected to the electric circuit to be protected. When detecting an abnormal current in the electric circuit by a built-in current sensor, an external current sensor connected to the electric circuit, or the like, the device S determines whether or not the abnormal current belongs to a relatively low current range lower than a predetermined value (for example, 1000 to 2000 A [amps]). When determining that the abnormal current belongs to the relatively low current range lower than the predetermined value, the device S inputs an abnormality signal X1 to the second power source **830**. Thereafter, after a predetermined time has elapsed, the device S inputs an abnormality signal X2 to the first power source P. Note that, as described later, the predetermined time is a time until the distal end portion **811** of the regulating body **810** projects into the housing portion **310** of the housing **300** by the second power source **830**. On the other hand, when determining that the detected abnormal current does not belong to the relatively low current range lower than the predetermined value and belongs to a relatively high current range higher than the predetermined value, the device S inputs the abnormality signal X2 only to the first power source P without inputting the abnormality signal X1 to the second power source **830**.

(60) Moreover, the paired electrodes **540** and **550** are arranged on the lower end side of the moving body **500** so as to face the cut portion **400**, and the insulator **560** away from the cut portion **400** is interposed between the paired electrodes and the cut portion **400**. Therefore, since the paired electrodes **540** and **550** are not physically and electrically connected to the cut portion **400**, the current flowing through the electric circuit does not flow in the fuse function portion **700** via the electrodes **540** and **550**. As a result, it is possible to prevent the current in the electric circuit from constantly flowing through the fuse function portion **700**, and it is possible to prevent heat generation and deterioration of the fuse function portion **700**. As described later, the electric circuit breaker **600** can induce an arc generated when the electric circuit is broken in the fuse function portion **700** to effectively and quickly extinguish the arc. Therefore, an arc-extinguishing material for extinguishing the arc is not enclosed in the housing portion **310** (in particular, around the cut piece **420**). Note that, basically, it is not necessary to enclose the arc-extinguishing material in the housing portion **310**, but the arc-extinguishing material may be enclosed in the housing portion **310** depending on the specification.

(61) Next, a state where the electric circuit breaker **600** breaks an electric circuit in a case where an overcurrent belonging to a relatively low current range lower than a predetermined value flows through the electric circuit will be described with reference to FIGS. **8** and **9**. Note that FIG. **8** is a cross-sectional view illustrating a state where the moving body **500** has moved from the state illustrated in FIG. **6**, and FIG. **9** is a cross-sectional view illustrating a state where the moving body **500** has moved from the state illustrated in FIG. **7**.

(62) First, assuming that, when detecting an abnormal current in the electric circuit, the device **S** determines that the abnormal current belongs to a relatively low current range lower than a predetermined value (for example, 1000 to 2000 A [amps]). Next, the device **S** inputs the abnormality signal **X1** to the second power source **830**. As a result, the gunpowder in the second power source **830** explodes, and the air pressure due to the explosion is transmitted to the terminal portion **812** of the regulating body **810**. Then, the regulating body **810** is forcefully blown toward the housing portion **310** of the housing **300** by the air pressure and instantaneously moves toward the moving body **500** in the housing portion **820** of the regulating unit **800**. As a result, as illustrated in FIG. **9**, the distal end portion **811** of the regulating body **810** projects into the housing portion **310** of the housing **300**.

(63) Thereafter, the device **S** inputs the abnormality signal **X2** to the first power source **P**. As a result, the gunpowder in the first power source **P** explodes, and the air pressure due to the explosion is transmitted to the recessed portion **511** on the upper end side of the moving body **500**. The moving body **500** is forcefully blown from the first end portion **320** toward the second end portion **330** by the air pressure, and instantaneously moves toward the second end portion **330** in the housing portion **310**.

(64) Then, as illustrated in FIGS. **8** and **9**, the cut piece **420** is strongly pushed downward by the insulator **560** of the moving body **500**, and the cut piece **420** is cut in the vicinity of the coupling portion between the cut piece **420** and the base piece **430** and physically separated from the base piece **430**. Therefore, the state where the base pieces **430** on both sides are energized is immediately broken, and an overcurrent can be prevented from flowing through the electric circuit. Note that, since the abnormal current belongs to the relatively low current range lower than the predetermined value, the arc discharge is not generated by the insulator **560** interposed between the base pieces **430** even if the distance between the cut piece **420** and the base piece **430B** that are separated is short.

(65) In addition, as illustrated in FIG. **8**, since the abutting portion **521** of the moving body **500** abuts on the distal end portion **811** of the regulating body **810** projecting into the housing portion **310**, the moving body **500** cannot further move toward the second end portion **330**. Therefore, the electrode **540** and the electrode **550** are not in contact with the base piece **430**, and the current flowing through the base piece **430** does not flow through the electrode **540** and the electrode **550** in the fuse function portion **700**. That is, the regulating unit **800** regulates the movement of the moving body **500** in a manner that a part of the cut portion **400** and the electrode do not come into contact with each other in order to make the cut portion **400** and the fuse function portion **700** unconnected. Note that the regulating unit **800** regulates the movement of the moving body **500** by causing the distal end portion **811** to abut on the abutting portion **521** of the moving body **500**, but is not limited thereto, and the regulating unit **800** may have any configuration as long as the movement of the moving body **500** can be regulated.

(66) Note that when the current belonging to the relatively low current range flows in the fuse function portion **700** through the electrode **540** and the electrode **550**, the current belongs to the relatively low current range, and thus the fusion portion **740** of the fuse function portion **700** is not fused and the current cannot be broken, or it takes a relatively long time to break the current, and the overcurrent flowing through the electric circuit cannot be broken immediately.

(67) Next, a state where the electric circuit breaker **600** breaks an electric circuit in a case where an overcurrent belonging to a relatively high current range higher than a predetermined value flows

through the electric circuit will be described with reference to FIGS. **10** to **12**. Note that FIGS. **10** to **12** are cross-sectional views illustrating a state where the moving body **500** has moved from the state illustrated in FIG. **6**.

(68) First, assuming that, when detecting an abnormal current in the electric circuit, the device **S** determines that the abnormal current does not belong to a relatively low current range lower than a predetermined value but belongs to a relatively high current range higher than the predetermined value. Next, the device **S** inputs the abnormality signal **X2** only to the first power source **P** without inputting the abnormality signal **X1** to the second power source **830**.

(69) As a result, the gunpowder in the first power source **P** explodes, and the moving body **500** instantaneously moves toward the second end portion **330** in the housing portion **310**. Then, as illustrated in FIG. **10**, the moving body **500** moves toward the second end portion **330**, the cut piece **420** is strongly pushed downward by the insulator **560** of the moving body **500**, and the cut piece **420** is cut in the vicinity of the coupling portion between the cut piece **420** and the base piece **430** and physically separated from the base piece **430**.

(70) In this state, since the electrode **540** and the electrode **550** are not in contact with the base piece **430**, the current **I1** flowing through the base piece **430** does not flow through the electrode **540** and the electrode **550** in the fuse function portion **700**. However, the cut piece **420** immediately after being cut and separated is close to the base piece **430**, and the abnormal current belongs to the relatively high current range higher than the predetermined value. Therefore, in this state, the arc discharge is instantaneously generated between the cut piece **420** and the base piece **430**, and the current **I1** can flow between the base pieces **430** on both sides through the cut piece **420**.

(71) Next, as illustrated in FIG. **11**, when the moving body **500** further moves toward the second end portion **330**, the electrode **540** and the electrode **550** come into contact with the base piece **430** in a state where the base piece **430** and the cut piece **420** remain energized by the arc discharge between the cut piece **420** and the base piece **430**. Then, the fuse function portion **700** is in a state of being energized with a part of the cut portion **400** via the electrode **540** and the electrode **550**, and a part **I2** of the current **I1** flowing through the electric circuit flows in the fuse function portion **700**. Note that, since the device **S** does not input the abnormality signal **X1** to the second power source **830**, the regulating unit **800** is not operated, and the distal end portion **811** of the regulating body **810** does not project into the housing portion **310** of the housing **300**. Therefore, the movement of the moving body **500** is not regulated by the regulating unit **800**.

(72) Next, as illustrated in FIG. **12**, when the moving body **500** further moves toward the second end portion **330**, the cut piece **420** is pushed and moved toward the second end portion **330** and is largely separated from the base piece **430**. Then, the arc discharge between the cut piece **420** and the base piece **430** is physically separated and disappears. Therefore, the state where the base pieces **430** on both sides of the cut portion **400** are energized via the cut piece **420** by the arc discharge is broken, and an overcurrent can be prevented from flowing through the electric circuit.

(73) In addition, as illustrated in FIG. **12**, when the cut piece **420** is largely separated from the base piece **430** and the energized state of the cut portion **400** is broken, the current **I1**(fault current) flowing through the electric circuit is induced in the fuse function portion **700**, so that it is possible to prevent the arc discharge between the cut piece **420** and the base piece **430** that are separated from being continuously generated. Note that, as illustrated in FIGS. **10** to **11**, the arc discharge generated immediately after the cut piece **420** is separated from the base piece **430** has little energy and disappears immediately because a part of the current **I1** is induced in the fuse function portion **700**. Therefore, even if the arc discharge is instantaneously generated immediately after the cut piece **420** is separated from the base piece **430**, the other parts of the electric circuit breaker **600** are not affected, and there is no problem in safety.

(74) As illustrated in FIG. **12**, the fusion portion **740** of the fuse function portion **700** is quickly fused by the current **I1** induced in the fuse function portion **700**, and the current flowing through the electric circuit is quickly broken. Furthermore, after the fusion portion **740** is fused, an arc is

generated between the terminals **750** of the fuse function portion **700** by the voltage applied to the base pieces **430** on both sides connected to the electric circuit, but the arc is quickly and effectively extinguished by the arc-extinguishing material **730** in the fuse function portion **700**.

(75) As described above, when the electric circuit breaker **600** breaks a relatively high current (fault current) flowing through the electric circuit, as illustrated in FIGS. **10** to **12**, in a state where the base pieces **430** on both sides of the cut portion **400** are energized by arc discharge via the cut piece **420**, the cut portion **400** is connected to the fuse function portion **700** via the pair of electrode **540** and electrode **550**, and thereafter, as illustrated in FIG. **12**, along with the movement of the moving body **500**, the cut piece **420** is largely separated from the base piece **430** to extinguish the arc discharge in a manner that the arc discharge does not continue any more, and a state where the base pieces **430** on both sides of the cut portion **400** are energized via the cut piece **420** is broken. That is, before the state where the cut portion **400** is energized is completely broken and the arc discharge is continuously generated between the base pieces **430** on both sides, the state where the cut portion **400** and the fuse function portion **700** are connected is secured, so that the arc due to a relatively high fault current can be reliably induced in the fuse function portion **700** and extinguished in the fuse function portion **700**. As a result, it is possible to prevent the electric circuit breaker **600** from being damaged by the arc between the base pieces **430** due to the fault current being continuously generated in the housing **300**, and to safely break the electric circuit.

(76) Note that, as illustrated in FIG. **12**, when the moving body **500** further moves toward the second end portion **330**, the cut piece **420** pushed out by the moving body **500** abuts on the abutment base **112**, and the moving body **500** stops. Since the insulator **560** is disposed between the base piece **430** and the cut piece **420**, between the electrode **540** and the cut piece **420**, and between the electrode **550** and the cut piece **420**, even if a voltage is inadvertently applied between the base pieces **430**, it is possible to prevent an arc from being generated between the cut piece **420** and the base piece **430** and the base pieces **430** on both sides from being energized. Furthermore, as illustrated in FIGS. **10** to **12**, after the pair of electrodes **540** and **550** come into contact with a part of the cut portion **400**, the electrodes **540** and **550** move toward the second end portion **330** and always maintain a state of being in contact with a part of the cut portion **400**, so that the state where the cut portion **400** is connected to the fuse function portion **700** is also always maintained.

(77) As described above, according to the electric circuit breaker **600** of the present invention, in a case where the overcurrent belonging to the relatively low current range flows through the electric circuit, as illustrated in FIGS. **8** and **9**, the cut portion **400** and the fuse function portion **700** are not connected, and the cut piece **420** between the base pieces **430** on both sides of the cut portion **400** is cut, and the state where the base pieces **430** on both sides are energized is immediately broken, so that the overcurrent is prevented from flowing through the electric circuit. Therefore, it is possible to solve the problem that, as in a conventional case, the current belonging to the relatively low current range cannot be broken because the fusion portion **740** of the fuse function portion **700** is not fused or the overcurrent flowing through the electric circuit cannot be broken immediately because it takes a relatively long time to break the current. On the other hand, in a case where the overcurrent belonging to the relatively high current range flows through the electric circuit, as illustrated in FIGS. **10** to **12**, the cut portion **400** and the fuse function portion **700** are connected, and the cut piece **420** between the base pieces **430** on both sides of the cut portion **400** is cut, and the state where the base pieces **430** on both sides are energized is immediately and safely broken, so that the overcurrent is prevented from flowing through the electric circuit. As described above, the electric circuit breaker **600** of the present invention quickly breaks current in a wide current range up to a relatively low current as well as a relatively high current.

Second Embodiment

(78) Next, an electric circuit breaker **600A** according to a second embodiment of the present invention will be described with reference to FIGS. **13** to **16**. In addition, since the configuration of the electric circuit breaker **600A** according to the second embodiment is basically the same as the

configuration of the electric circuit breaker **600** according to the first embodiment except for the configuration of a regulating unit **800A**, the description of the same configuration will be omitted. Note that FIG. **13(a)** is an overall perspective view of a lower housing **100A**, FIG. **13(b)** is a plan view of the lower housing **100A**, and FIG. **13(c)** is a cross-sectional view taken along line G-G.

(79) The regulating unit **800A** formed of an insulator such as a synthetic resin is attached to the lower housing **100A**. The regulating unit **800A** includes a housing portion **820A** in which a regulating body **810A** is slidably housed, and a second power source **830A** for moving the regulating body **810A**. The regulating body **810A** has a substantially rectangular parallelepiped shape, includes a terminal portion **812A** and a distal end portion **811A**, and includes a space **813A** in which a part of an abutment base **112A** to be described later can be housed. In addition, the lower housing **100A** includes a through-hole **150A** that allows a lower housing portion **110A** in the lower housing and the housing portion **820A** of the regulating unit **800A** to communicate with each other, and the through-hole **150A** is formed in a manner that the regulating body **810A** of the regulating unit **800A** can be inserted therethrough. As described in detail later, power such as air pressure generated from the second power source **830A** housed in the housing portion **820A** is transmitted to the regulating body **810A** in the housing portion **820A**, and the moving regulating body **810A** is inserted through the through-hole **150A** and moved to the lower housing portion **110A** of the lower housing **100A**. Note that the regulating unit **800A** is attached to the lower housing **100A**, but is not limited thereto, and can be attached to any place as long as it is a part of the housing **300**.

(80) Next, a method of assembling the electric circuit breaker **600A** of the present invention will be described with reference to FIG. **14**. Note that FIG. **14** is an exploded perspective view of the electric circuit breaker **600A**.

(81) When the electric circuit breaker **600A** is assembled, first, the abutment base **112A** with a substantially T shape formed of an insulator is fixed to the bottom of the lower housing portion **110A** of the lower housing **100A**. Next, a cut portion **400A** is disposed in a manner that a cut piece **420A** traverses the lower housing portion **110A** of the lower housing **100A**.

(82) Next, an upper housing **200A** is fitted from above the lower housing **100A** in a manner that the side of a body **510A** of a moving body **500A** is inserted into an upper housing portion **210A** of the upper housing **200A**. Then, a housing **300A** including the lower housing **100A** and the upper housing **200A** is assembled in a state where the cut portion **400A** and the moving body **500A** are housed therein. Furthermore, a first power source PA is attached to a power source storage portion **221A** of the upper housing **200A**, and a part of the first power source PA is housed in a recessed portion **511A** of the moving body **500A**.

(83) In addition, the regulating unit **800A** is attached to the lower housing **100A**. A part of the regulating body **810A** is configured to be movable into the lower housing portion **110A** of the lower housing **100A** by the second power source **830A**. When it is detected that an abnormal current flows through the electric circuit and an abnormality signal is input from an external device to the second power source **830A**, for example, the gunpowder in the second power source **830A** is exploded, and the regulating body **810A** is instantaneously pushed out and moved in the housing portion **820A** by the air pressure due to the explosion.

(84) The electric circuit breaker **600A** also includes a fuse function portion **700A**. Terminals **750A** on both sides of a fuse element **720A** are electrically connected to paired electrodes **540A** and **550A** by connection members **760A** such as electric wires. Note that the fuse function portion **700A** can be attached to any place in the housing **300A**.

(85) Next, an internal structure of the electric circuit breaker **600A** according to the second embodiment of the present invention will be described with reference to FIG. **15**. FIG. **15** is a cross-sectional view taken along line H-H in a state where the electric circuit breaker **600A** illustrated in FIG. **14** is assembled.

(86) The assembled and completed electric circuit breaker **600A** is attached in an electric circuit to

be protected and used. Specifically, a base piece **430A** of the cut portion **400A** is connected to a part of the electric circuit, and the cut portion **400A** constitutes a part of the electric circuit. An insulator **560A** provided at the lower end of the moving body **500A** extends along the cut piece **420A** and is disposed away from the cut piece **420A**. In a normal state, since the base piece **430A** and the cut piece **420A** of the cut portion **400A** are not cut and are physically and electrically connected, a current flows through the electric circuit via the base piece **430A** and the cut piece **420A** of the cut portion **400A**. In addition, the side of the distal end portion **811A** of the regulating body **810A** of the regulating unit **800A** is inserted into the through-hole **150A** of the housing **300A** but does not project to the housing portion **310A**. Therefore, in the normal state, a part of the regulating body **810A** of the regulating unit **800A** does not project to the housing portion **310A**, and the regulating unit **800A** does not regulate the movement of the moving body **500A**.

(87) Furthermore, as illustrated in FIG. **15**, a device SA that detects an abnormal current in the electric circuit is connected to the electric circuit to be protected. When determining that the abnormal current belongs to a relatively low current range lower than a predetermined value, the device SA inputs an abnormality signal X1A to the second power source **830A**. Thereafter, after a predetermined time has elapsed, the device SA inputs an abnormality signal X2A to the first power source PA. As described later, the predetermined time is a time until the regulating body **810A** projects into the housing portion **310A** of the housing **300A** by the second power source **830A** and the regulating unit **800A** is brought to a state capable of regulating the movement of the moving body **500A**. On the other hand, when determining that the detected abnormal current does not belong to the relatively low current range lower than the predetermined value and belongs to a relatively high current range higher than the predetermined value, the device SA inputs the abnormality signal X2A only to the first power source PA without inputting the abnormality signal X1A to the second power source **830A**.

(88) Next, a state where the electric circuit breaker **600A** breaks an electric circuit in a case where an overcurrent belonging to a relatively low current range lower than a predetermined value flows through the electric circuit will be described with reference to FIG. **16**. Note that FIG. **16** is a cross-sectional view illustrating a state where the moving body **500A** has moved from the state illustrated in FIG. **15**.

(89) First, assuming that, when detecting an abnormal current in the electric circuit, the device SA determines that the abnormal current belongs to a relatively low current range lower than a predetermined value (for example, 1000 to 2000 A [amps]). Next, the device SA inputs the abnormality signal X1A to the second power source **830A**. As a result, the gunpowder in the second power source **830A** explodes, and the air pressure due to the explosion is transmitted to the terminal portion **812A** of the regulating body **810A**. Then, the regulating body **810A** is forcefully blown toward the housing portion **310A** of the housing **300A** by the air pressure and moves in the housing portion **820A**. The regulating body **810A** then projects into the housing portion **310A** of the housing **300A** and is located under the moving body **500**. At this time, since the abutment base **112A** disposed in the housing portion **310A** is housed in the space **813A** of the regulating body **810A**, it does not interfere with the movement of the regulating body **810A**.

(90) Thereafter, the device SA inputs the abnormality signal X2A to the first power source PA. As a result, the gunpowder in the first power source PA explodes, and the moving body **500A** is forcefully blown from the first end portion **320A** toward the second end portion **330A** by the air pressure due to the explosion, and instantaneously moves toward the second end portion **330A** in the housing portion **310A**.

(91) Then, the cut piece **420A** is strongly pushed downward by the insulator **560A** of the moving body **500A**, and the cut piece **420A** is cut in the vicinity of the coupling portion between the cut piece **420A** and the base piece **430A** and physically separated from the base piece **430A**. Therefore, the state where the base pieces **430A** on both sides are energized is immediately broken, and an overcurrent can be prevented from flowing through the electric circuit. Note that, since the

abnormal current belongs to the relatively low current range lower than the predetermined value, the arc discharge is not generated even if the distance between the cut piece **420A** and the base piece **430A** that are separated is short, and it is possible to more reliably prevent the arc discharge from being generated by the insulator **560A** interposed between the base pieces **430A**.

(92) In addition, since the lower end side of the moving body **500A** abuts on the regulating body **810A** projecting into the housing portion **310A** so as to sandwich the separated cut piece **420A** with the regulating body, the moving body **500A** cannot further move toward the second end portion **330A**. In this state, similarly to the electric circuit breaker **600** according to the first embodiment illustrated in FIG. **8**, since the electrode is not in contact with the base piece **430A**, the current flowing through the base piece **430A** does not flow through the electrode in the fuse function portion **700A**. That is, the regulating body **810A** regulates the movement of the moving body **500A** in a manner that a part of the cut portion **400A** and the electrode do not come into contact with each other in order to make the cut portion **400A** and the fuse function portion **700A** unconnected. Note that the regulating body **810A** of the regulating unit **800A** abuts on the moving body **500A** so as to be located under the moving body **500A**, and regulates the movement of the moving body **500A**. Therefore, the regulating body **810A** of the regulating unit **800A** can receive the lower side of the moving body **500A** in a firm and stable state, and the structure of the regulating body **810A** of the regulating unit **800A** is also simplified.

(93) Next, the case where an overcurrent belonging to a relatively high current range higher than a predetermined value flows through the electric circuit will be described. When detecting an abnormal current in the electric circuit, the device SA determines that the abnormal current does not belong to a relatively low current range lower than a predetermined value but belongs to a relatively high current range higher than the predetermined value. Next, the device SA inputs the abnormality signal X2A only to the first power source PA without inputting the abnormality signal X1A to the second power source **830A**. Note that, since the device SA does not input the abnormality signal X1A to the second power source **830A**, the regulating unit **800A** is not operated, and the regulating body **810A** does not project into the housing portion **310A** of the housing **300A**, and thus the movement of the moving body **500A** is not regulated by the regulating unit **800A**.

(94) Then, the gunpowder in the first power source P explodes by the abnormality signal X2A, and the moving body **500A** instantaneously moves toward the second end portion **330A** in the housing portion **310A**. The subsequent operation of the electric circuit breaker **600A** to break the electric circuit is similar to the operation mode of the electric circuit breaker **600** according to the first embodiment illustrated in FIGS. **10** to **12**. As a result, it is possible to prevent the electric circuit breaker **600A** from being damaged by the arc between the base pieces **430A** due to a fault current being continuously generated in the housing **300A**, and to safely break the electric circuit.

Third Embodiment

(95) Next, an electric circuit breaker **600B** according to a third embodiment of the present invention will be described with reference to FIGS. **17** to **25**. In addition, since the configuration of the electric circuit breaker **600B** according to the third embodiment is basically the same as the configuration of the electric circuit breaker **600** according to the first embodiment except that the arrangement of an electrode **540B** and an electrode **550B**, and a conductor **570B** is provided, the description of the same configuration will be omitted. Note that FIG. **17(a)** is an overall perspective view of a lower housing **100B**, FIG. **17(b)** is a plan view of the lower housing **100B**, and FIG. **17(c)** is a cross-sectional view taken along line I-I.

(96) A regulating unit **800B** formed of an insulator such as a synthetic resin is attached to the lower housing **100B**. The regulating unit **800B** includes a housing portion **820B** in which a regulating body **810B** is slidably housed, and a second power source **830B** for moving the regulating body **810B**. The regulating body **810B** has a substantially rectangular parallelepiped shape with a sharp upper end **814B**, includes a terminal portion **812B** and a distal end portion **811B**, and includes a substantially rectangular parallelepiped space **813B** in which a part of an abutment base **112B** to be

described later can be housed. In addition, the lower housing **100B** includes a through-hole **150B** that allows a lower housing portion **110B** in the lower housing and the housing portion **820B** of the regulating unit **800B** to communicate with each other, and the through-hole **150B** is formed in a manner that the regulating body **810B** of the regulating unit **800B** can be inserted therethrough. As described in detail later, power such as air pressure generated from the second power source **830B** housed in the housing portion **820B** is transmitted to the regulating body **810B** in the housing portion **820B** to move the regulating body **810B** toward the lower housing portion **110B** of the lower housing **100B**. Then, the regulating body **810B**, which has moved, passes through the through-hole **150B** and moves into the lower housing portion **110B** of the lower housing **100B**. Note that the regulating unit **800B** is attached to the lower housing **100B**, but is not limited thereto, and can be attached to any place as long as it is a part of the housing **300B**.

(97) Next, a method of assembling the electric circuit breaker **600B** of the present invention will be described with reference to FIG. **18**. Note that FIG. **18** is an exploded perspective view of the electric circuit breaker **600B**.

(98) When the electric circuit breaker **600B** is assembled, first, the substantially rectangular parallelepiped abutment base **112B** with a sharp distal end **118B** formed of an insulator is fixed to the bottom of the lower housing portion **110B** of the lower housing **100B**. Next, a cut portion **400B** is disposed in a manner that a cut piece **420B** traverses the lower housing portion **110B** of the lower housing **100B**.

(99) Next, an upper housing **200B** is fitted from above the lower housing **100B** in a manner that the side of a body **510B** of a moving body **500B** is inserted into an upper housing portion **210B** of the upper housing **200B**. Then, a housing **300B** including the lower housing **100B** and the upper housing **200B** is assembled in a state where the cut portion **400B** and the moving body **500B** are housed therein. Furthermore, a first power source PB is attached to a power source storage portion **221B** of the upper housing **200B**, and a part of the first power source PB is housed in a recessed portion **511B** of the moving body **500B**.

(100) In addition, the regulating unit **800B** is attached to the lower housing **100B**. A part of the regulating body **810B** is configured to be movable into the lower housing portion **110B** of the lower housing **100B** by the second power source **830B**. When it is detected that an abnormal current flows through the electric circuit and an abnormality signal is input from an external device to the second power source **830B**, for example, the gunpowder in the second power source **830B** is exploded, and the regulating body **810B** is instantaneously pushed out and moved in the housing portion **820B** by the air pressure due to the explosion.

(101) The electric circuit breaker **600B** also includes a fuse function portion **700B**. Terminals **750B** on both sides of a fuse element **720B** are electrically connected to paired electrodes **540B** and **550B** arranged in the lower housing portion **110B** of the lower housing **100B** by connection members **760B** such as electric wires. Note that the fuse function portion **700B** can be attached to any place in the housing **300B**.

(102) Next, an internal structure of the electric circuit breaker **600B** according to the third embodiment of the present invention will be described with reference to FIGS. **19** and **20**. Note that FIG. **19** is a cross-sectional view taken along line J-J in a state where the electric circuit breaker **600B** illustrated in FIG. **18** is assembled, and FIG. **20** is a cross-sectional view taken along line K-K in a state where the electric circuit breaker **600B** illustrated in FIG. **18** is assembled.

(103) The assembled and completed electric circuit breaker **600B** is attached in an electric circuit to be protected and used. Specifically, a base piece **430B** of the cut portion **400B** is connected to a part of the electric circuit, and the cut portion **400B** constitutes a part of the electric circuit. Furthermore, the electrode **540B** and the electrode **550B** are arranged on the side of a second end portion **330B** in the housing portion **310B** of the housing **300B**, and are located on the opposite side of the moving body **500B** with the cut piece **420B** interposed therebetween. In addition, the fuse function portion **700B** is fixed at an arbitrary position of the housing **300B**. Furthermore, a

pair of conductors **570B** formed of metal such as copper is provided on the distal end side of the moving body **500B** so as to face the cut pieces **420B**. Note that, in a normal state, since the base piece **430B** and the cut piece **420B** of the cut portion **400B** are not cut and are physically and electrically connected, a current **I1B** flows through the electric circuit via the base piece **430B** and the cut piece **420B** of the cut portion **400B**. Note that the paired electrodes **540B** and **550B** are arranged below the cut piece **420B** away from the cut piece **420B**. Therefore, since the paired electrodes **540B** and **550B** are not physically and electrically connected to the cut portion **400B**, the current flowing through the electric circuit does not flow in the fuse function portion **700B** via the electrodes **540B** and **550B**. In addition, the conductors **570B** on both sides are physically separated from each other and are not electrically connected to each other. Moreover, the conductor **570B** is disposed above the cut piece **420B** away from the cut piece **420B**.

(104) In addition, as illustrated in FIG. **20**, the side of the distal end portion **811B** of the regulating body **810B** of the regulating unit **800B** is inserted into the through-hole **150B** of the housing **300B** but does not project to the housing portion **310B**. Therefore, in the normal state, a part of the regulating body **810B** of the regulating unit **800B** does not project to the housing portion **310B**, and the regulating unit **800B** does not regulate the movement of the moving body **500B**.

(105) Furthermore, as illustrated in FIG. **20**, a device **SB** that detects an abnormal current in the electric circuit is connected to the electric circuit to be protected. When determining that the abnormal current belongs to a relatively low current range lower than a predetermined value, the device **SB** inputs an abnormality signal **X1B** to the second power source **830B**. Thereafter, after a predetermined time has elapsed, the device **SB** inputs an abnormality signal **X2B** to the first power source **PB**. Note that, as described later, the predetermined time is a time until the regulating body **810B** projects into the housing portion **310B** of the housing **300B** by the second power source **830B**. On the other hand, when determining that the detected abnormal current does not belong to the relatively low current range lower than the predetermined value and belongs to a relatively high current range higher than the predetermined value, the device **SB** inputs the abnormality signal **X2B** only to the first power source **PB** without inputting the abnormality signal **X1B** to the second power source **830B**.

(106) Next, a state where the electric circuit breaker **600B** breaks an electric circuit in a case where an overcurrent belonging to a relatively low current range lower than a predetermined value flows through the electric circuit will be described with reference to FIGS. **21** and **22**. Note that FIG. **21** is a cross-sectional view illustrating a state where the moving body **500B** has moved from the state illustrated in FIG. **19**, and FIG. **22** is a cross-sectional view illustrating a state where the moving body **500B** has moved from the state illustrated in FIG. **20**.

(107) First, when detecting an abnormal current in the electric circuit, the device **SB** determines that the abnormal current belongs to a relatively low current range lower than a predetermined value (for example, 1000 to 2000 A [amps]). Next, the device **SB** inputs the abnormality signal **X1B** to the second power source **830B**. As a result, the gunpowder in the second power source **830B** explodes, and the air pressure due to the explosion is transmitted to the terminal portion **812B** of the regulating body **810B**. Then, the regulating body **810B** is forcefully blown toward the housing portion **310B** of the housing **300B** by the air pressure and moves in the housing portion **820B**. The regulating body **810B** then projects into the housing portion **310B** of the housing **300B** and is located under the moving body **500B**. At this time, since the abutment base **112B** disposed in the housing portion **310B** is housed in the space **813B** of the regulating body **810B**, it does not interfere with the movement of the regulating body **810B**. A height **L1** of the regulating body **810B** is higher than a height **L2** of the abutment base **112B**. Therefore, as described later, by the moving body **500B** abutting on the regulating body **810B** with a high height, the amount of movement by which the moving body **500B** can move downward toward the second end portion **330B** can be further regulated.

(108) Thereafter, the device **SB** inputs the abnormality signal **X2B** to the first power source **PB**. As

a result, the gunpowder in the first power source PB explodes, and the moving body 500B is forcefully blown from the first end portion 320B toward the second end portion 330B by the air pressure due to the explosion, and instantaneously moves toward the second end portion 330B in the housing portion 310B.

(109) As illustrated in FIGS. 21 and 22, the cut piece 420B is then strongly pushed downward by the moving body 500B. Then, the cut piece 420B is bent in a substantially L shape by the sharp upper end 814B of the regulating body 810B and cut in the vicinity of the coupling portion between the cut piece 420B and the base piece 430B to be physically separated from the base piece 430B. Therefore, the state where the base pieces 430B on both sides are energized is immediately broken, and an overcurrent can be prevented from flowing through the electric circuit. Note that, since the abnormal current belongs to the relatively low current range lower than the predetermined value, the arc is not generated through the cut piece 420B even if the distance between the cut piece 420B and the base piece 430B that are separated is short.

(110) In addition, since a projection 530B of the moving body 500B abuts on the upper end 814B of the regulating body 810B projecting into the housing portion 310B, the moving body 500B cannot further move toward the second end portion 330B. Then, the conductors 570B of the moving body 500B are not in contact with the electrode 540B and the electrode 550B, respectively. Note that the cut piece 420B, which has been cut, is also not in contact with the electrodes 540B and the electrode 550B.

(111) Therefore, since the electrode 540B and the electrode 550B are not electrically connected to the individual base pieces 430 via the conductors 570B, the current flowing through the base piece 430B does not flow through the electrode and the conductor 570B in the fuse function portion 700B. That is, the regulating body 810B regulates the movement of the moving body 500B in a manner that a part of the cut portion 400 and the electrode do not come into contact with each other in order to make the cut portion 400B and the fuse function portion 700B unconnected.

(112) Next, a state where the electric circuit breaker 600B breaks an electric circuit in a case where an overcurrent belonging to a relatively high current range higher than a predetermined value flows through the electric circuit will be described with reference to FIGS. 23 to 25. Note that FIGS. 23 to 25 are cross-sectional views illustrating a state where the moving body 500B has moved from the state illustrated in FIG. 19.

(113) First, when detecting an abnormal current in the electric circuit, the device SB determines that the abnormal current does not belong to a relatively low current range lower than a predetermined value but belongs to a relatively high current range higher than the predetermined value. Next, the device SB inputs the abnormality signal X2B only to the first power source PB without inputting the abnormality signal X1B to the second power source 830B.

(114) As a result, the gunpowder in the first power source PB explodes, and the moving body 500B instantaneously moves toward the second end portion 330B in the housing portion 310B. Then, the paired conductors 570B arranged on the lower end side of the moving body 500B come into contact with the cut piece 420B of the cut portion 400B. Then, as illustrated in FIG. 23, when the moving body 500B further moves toward the second end portion 330B, the cut piece 420B is strongly pushed downward by the conductor 570B and the projection 530B of the moving body 500B, and the cut piece 420B is cut in the vicinity of the coupling portion between the cut piece 420B and the base piece 430B and physically separated from the base piece 430B. Note that, since the conductor 570B is in contact with the cut piece 420B and the base piece 430B, the cut piece 420B is physically separated from the base piece 430B, but the conductor 570B keeps the base pieces 430B on both sides of the cut portion 400B energized via the cut piece 420B.

(115) When the moving body 500B further moves toward the second end portion 330B, as illustrated in FIG. 24, the conductors 570B on both sides come into contact with the electrode 540B and the electrode 550B, respectively. The conductor 570B is also in contact with the base piece 430B. Therefore, the fuse function portion 700B is in a state of being energized with a part of the

cut portion **400B** via the conductor **570B** and the pair of electrodes (**540B**, **550B**), and a part **I2B** of the current flowing through the electric circuit flows in the fuse function portion **700B**. Furthermore, in the state illustrated in FIG. **24**, since the cut piece **420B** is in contact with the conductor **570B**, the cut piece is electrically connected to the base piece **430B** via the conductor **570B**. That is, in a state where the base pieces **430B** on both sides of the cut portion **400B** are energized via the cut piece **420B**, a part of the cut portion **400B** is connected to the fuse function portion **700B**.

(116) Next, as illustrated in FIG. **25**, when the moving body **500B** further moves toward the second end portion **330B**, the cut piece **420B** is strongly pushed downward by the projection **530B** and the conductor **570B** of the moving body **500B**, and the cut piece **420B** is bent in a substantially L shape by the triangular distal end **118B** of the abutment base **112B**. Therefore, the cut piece **420B** and the conductor **570B** are separated from each other, and the cut piece **420B** and the conductor **570B** are not physically and electrically connected to each other. That is, the state where the base pieces **430B** on both sides of the cut portion **400B** are energized via the cut piece **420B** is broken, and an overcurrent can be prevented from flowing through the electric circuit.

(117) In addition, as illustrated in FIGS. **24** to **25**, after the paired electrodes **540B** and **550B** come into contact with a part of the cut portion **400B** via the conductor **570B** and the cut portion **400B** is connected to the fuse function portion **700B**, the cut piece **420B** as a part of the cut portion **400B** is bent and the state where the base pieces **430B** on both sides of the cut portion **400B** are energized via the cut piece **420B** is broken. Therefore, when the energized state of the cut portion **400B** is broken, a current **I1B** (fault current) flowing through the base piece **430B** is induced in the fuse function portion **700B**. Therefore, it is possible to prevent the arc due to the fault current from being generated between the cut piece **420B** and the base piece **430B** that are divided.

(118) As illustrated in FIG. **25**, a fusion portion **740B** of the fuse function portion **700B** is quickly fused by the current **I1B** induced in the fuse function portion **700B**, and the current flowing through the electric circuit is quickly broken. Furthermore, after the fusion portion **740B** is fused, an arc is generated between the terminals **750B** of the fuse function portion **700B** by the voltage applied to the base pieces **430B** on both sides connected to the electric circuit, but the arc is quickly and effectively extinguished by an arc-extinguishing material **730B** in the fuse function portion **700B**. Note that, as illustrated in FIGS. **24** to **25**, after the pair of conductors **570B** comes into contact with a part of the cut portion **400B** and the pair of electrodes (**540B**, **550B**), the conductor **570B** moves toward the second end portion **330B** and always maintain a state of being in contact with a part of the cut portion **400B** and the pair of electrodes (**540B**, **550B**), so that the state where the cut portion **400B** is connected to the fuse function portion **700B** is also always maintained.

(119) As described above, in the electric circuit breaker **600B**, a relatively high current (fault current) flowing through the electric circuit when the electric circuit is broken is induced in the fuse function portion **700B**, and the arc generated by the induced current can be effectively and quickly extinguished in the fuse function portion **700B**. In addition, before the state where the cut portion **400B** is energized is broken and an arc is generated between the base pieces **430B** on both sides, the state where the cut portion **400B** and the fuse function portion **700B** are connected is secured, so that the arc due to a relatively high fault current can be reliably induced in the fuse function portion **700B** and extinguished in the fuse function portion **700B**. As a result, it is possible to prevent the electric circuit breaker **600B** from being damaged by the arc generated between the base pieces **430B** in the housing **300B**, and to safely break the electric circuit.

(120) Furthermore, by providing the pair of electrodes (**540B**, **550B**) and the fuse function portion **700B** not on the side of the moving body **500B** but on the side of the housing **300B**, it is possible to easily maintain a state where the connectivity between the pair of electrodes (**540B**, **550B**) and the fuse function portion **700B** is stably and reliably kept without being affected by the movement of the moving body **500B**. Therefore, the connection configuration (connection member or the like) between the pair of electrodes (**540B**, **550B**) and the fuse function portion **700B** can be simplified

without considering the movement of the moving body **500B**.

(121) As described above, according to the electric circuit breaker **600B** of the present invention, in a case where the overcurrent belonging to the relatively low current range flows through the electric circuit, as illustrated in FIGS. **21** and **22**, under a state where the cut portion **400B** and the fuse function portion **700B** are not connected, the cut piece **420B** between the base pieces **430B** on both sides of the cut portion **400B** is cut and the state where the base pieces **430B** on both sides are energized is immediately broken, so that the overcurrent is prevented from flowing through the electric circuit. Therefore, it is possible to solve the problem that, as in a conventional case, the current belonging to the relatively low current range cannot be broken because the fusion portion **740B** of the fuse function portion **700B** is not fused or the overcurrent flowing through the electric circuit cannot be broken immediately because it takes a relatively long time to break the current. On the other hand, in a case where the overcurrent belonging to the relatively high current range flows through the electric circuit, as illustrated in FIGS. **23** to **25**, under a state where the cut portion **400B** and the fuse function portion **700B** are connected, the cut piece **420B** between the base pieces **430B** on both sides of the cut portion **400B** is cut and the state where the base pieces **430B** on both sides are energized is immediately and safely broken, so that the overcurrent is prevented from flowing through the electric circuit. As described above, the electric circuit breaker **600B** of the present invention quickly breaks current in a wide current range up to a relatively low current as well as a relatively high current.

Fourth Embodiment

(122) Next, an electric circuit breaker **600C** according to a fourth embodiment of the present invention will be described with reference to FIGS. **26** and **27**. In addition, since the configuration of the electric circuit breaker **600C** according to the fourth embodiment is basically the same as the configuration of the electric circuit breaker **600** according to the first embodiment except that the regulating unit **800** is not provided, and a circuit **900C** and a breaker **970C** is provided, the description of the same configuration will be omitted. Note that FIG. **26** is an overall perspective view illustrating the electric circuit breaker **600C** in an exploded manner, FIG. **27(a)** is a cross-sectional view taken along line S-S of FIG. **26**, and FIG. **27(b)** is a cross-sectional view taken along line L-L of FIG. **26**.

(123) As illustrated in FIGS. **26** and **27**, a lower housing **100C** is a substantially quadrangular prism formed of an insulator such as a synthetic resin, and includes a hollow lower housing portion **110C** therein. The lower housing portion **110C** is configured to house a moving body **500C**. In addition, the lower housing **100C** includes a hollow lower housing portion **160C** so as to be adjacent to the lower housing portion **110C**. The lower housing portion **160C** is configured to house the breaker **970C**.

(124) Furthermore, placement portions **113C** recessed based on the shape of a base piece **430C** are provided in a part of an upper surface **120C** of the lower housing **100C** so that the base piece **430C** of a cut portion **400C** can be placed. The placement portions **113C** are arranged on both sides of the lower housing portion **110C** so as to face each other, and support the cut portion **400C** extending linearly on both sides.

(125) In addition, the circuit **900C** is connected in parallel with the cut portion **400C**. The entire circuit **900C** is a metal conductor such as copper in order to be electrically connected to the cut portion **400C** via a fuse function portion **700C**. The circuit **900C** includes a base piece **930C** directly coupled to one base piece **430C** of the cut portion **400C** and a base piece **930C** coupled to the other base piece **430C** of the cut portion **400C** via the fuse function portion **700C**. Moreover, the circuit includes a cut piece **940C** positioned between the base pieces **930C**. Furthermore, placement portions **115C** recessed based on the shape of the base piece **930C** are provided in a part of the upper surface **120C** of the lower housing **100C** so that the base piece **930C** of the circuit **900C** can be placed. The placement portions **115C** are arranged on both sides of the lower housing portion **160C** so as to face each other, and support the circuit **900C** extending linearly on both

sides.

(126) Furthermore, an upper housing **200C** is a substantially quadrangular prism formed of an insulator such as a synthetic resin, and forms a pair with the lower housing **100C** to constitute a housing **300C**. The upper housing includes a hollow upper housing portion **210C** therein, and the upper housing portion **210C** is configured to house the moving body **500C**. In addition, the upper housing **200C** also includes a hollow upper housing portion **170C** so as to be adjacent to the upper housing portion **210C**. The upper housing portion **170C** is configured to house the breaker **970C**.

(127) Furthermore, insertion portions **213C** recessed based on the shape of the base piece **430C** are provided in a part of a lower surface **230C** of the upper housing **200C** so that the base piece **430C** of the cut portion **400C** can be placed. The insertion portions **213C** are arranged on both sides of the upper housing portion **210C** so as to face each other, and are arranged at positions corresponding to the placement portions **113C** of the lower housing **100C**. Furthermore, insertion portions **215C** recessed based on the shape of the base piece **930C** are provided in a part of the lower surface **230C** of the upper housing **200C** so that the base piece **930C** of the circuit **900C** can be placed. The insertion portions **215C** are arranged on both sides of the upper housing portion **170C** so as to face each other, and support the circuit **900C** extending linearly on both sides.

(128) Moreover, a power source storage portion **221C** in which a first power source **PC** is housed is formed in a part of the side of an upper surface **220C** of the upper housing **200C**. The power source storage portion **221C** communicates with the upper end side of the upper housing portion **210C**. When it is detected that an abnormal current flows through the electric circuit, an abnormality signal is input from an external device to the first power source **PC**. Then, for example, the gunpowder in the first power source **PC** is exploded, and the moving body **500C** is instantaneously pushed out and moved in a housing portion **310C** including the upper housing portion **210C** and the lower housing portion **110C** by the air pressure due to the explosion. Note that the housing portion **310C** extends from a first end portion **320C** of the housing **300C** to a second end portion **330C** opposite to the first end portion **320C**. Since the moving body **500C** is disposed on the side of the first end portion **320C**, the moving body can move toward the second end portion **330C** in the housing portion **310C** by the first power source **PC** provided on the side of the first end portion **320C**.

(129) Moreover, a power source storage portion **241C** in which a second power source **990C** is housed is formed in a part of the side of the upper surface **220C** of the upper housing **200C**. The power source storage portion **241C** communicates with the upper end side of the upper housing portion **170C**. When it is detected that an abnormal current flows through the electric circuit, an abnormality signal is input from an external device to the second power source **990C**. Then, for example, the gunpowder in the second power source **990C** is exploded, and the breaker **970C** is instantaneously pushed out and moved in a housing portion **380C** including the upper housing portion **170C** and the lower housing portion **160C** by the air pressure due to the explosion. Note that the housing portion **380C** extends from the first end portion **320C** of the housing **300C** to the second end portion **330C** opposite to the first end portion **320C**. Since the breaker **970C** is disposed on the side of the first end portion **320C**, the breaker can move toward the second end portion **330C** in the housing portion **380C** by the second power source **990C** provided on the side of the first end portion **320C**.

(130) The electric circuit breaker **600C** also includes the fuse function portion **700C**. The fuse function portion **700C** includes a fuse element **720C** made of a conductive metal such as copper or an alloy thereof in a hollow and insulating casing **710C**, and the periphery of the fuse element **720C** inside the casing **710C** is filled with an arc-extinguishing material **730C**. One terminal **750C** of the fuse function portion **700C** is connected to the base piece **430C** of the cut portion **400C**, and the other terminal **750C** of the fuse function portion **700C** is connected to the base piece **930C** of the circuit **900C**. Therefore, the fuse function portion **700C** is electrically connected to the cut portion **400C** via the circuit **900C**. In addition, the fuse element **720C** includes a fusion portion

740C between the terminals **750C**, and the fusion portion **740C** is a portion in which the width of the fuse element **720C** is locally narrowed, and is configured to generate heat and fuse to break the current when the current to be broken by the electric circuit breaker flows. Note that the fuse function portion **700C** is housed in a housing portion **251C** of the upper housing **200C**.

(131) Further, as illustrated in FIG. 27, the electric circuit breaker **600C** is attached in an electric circuit to be protected and used. Specifically, the base piece **430C** of the cut portion **400C** is connected to a part of the electric circuit, and the cut portion **400C** constitutes a part of the electric circuit. In addition, a projection **530C** of the moving body **500C** extends along the cut piece **420C** and is disposed away from the cut piece **420C**. In a normal state, since the base piece **430C** and the cut piece **420C** of the cut portion **400C** are not cut and are physically and electrically connected, a current **I1C** flows through the electric circuit via the base piece **430C** and the cut piece **420C** of the cut portion **400C**.

(132) In addition, in the normal state, a projection **971C** of the breaker **970C** extends along the cut piece **940C** and is disposed away from the cut piece **940C**. That is, the circuit **900C** is neither disconnected nor broken by the breaker **970C**. Note that the resistance value of the fuse function portion **700C** is larger than the resistance value of the cut portion **400C**. Since the current **I1C** flowing through the cut portion **400C** and a current **I1C'** flowing through the fuse element **720C** are proportional to the reciprocal of each resistance value, the magnitude of the current **I1C'** in the normal state is as small as about ten percent of the total current (current **I1C**+current **I1C'**).

(133) Furthermore, as illustrated in FIG. 27, a device **SC** that detects an abnormal current in the electric circuit is connected to the electric circuit to be protected. When detecting an abnormal current in the electric circuit by a built-in current sensor, an external current sensor connected to the electric circuit, or the like, the device **SC** determines whether or not the abnormal current belongs to a relatively low current range lower than a predetermined value (for example, 1000 to 2000 A [amps]). When determining that the abnormal current belongs to the relatively low current range lower than the predetermined value, the device **SC** inputs an abnormality signal **X1C** to the second power source **990C**. Thereafter, after a predetermined time has elapsed, the device **SC** inputs an abnormality signal **X2C** to the first power source **PC**. Note that, as described later, the predetermined time is a time until the breaker **970C** cuts the cut piece **940C** of the circuit **900C** by the second power source **990C**. On the other hand, when determining that the detected abnormal current does not belong to the relatively low current range lower than the predetermined value and belongs to a relatively high current range higher than the predetermined value, the device **SC** inputs the abnormality signal **X2C** only to the first power source **PC** without inputting the abnormality signal **X1C** to the second power source **990C**.

(134) Note that, as described later, in a case where a relatively high current flows, the electric circuit breaker **600C** can induce an arc generated when the electric circuit is broken in the fuse function portion **700C** to effectively and quickly extinguish the arc. Therefore, an arc-extinguishing material for extinguishing the arc is not enclosed in the housing portion **310C** (in particular, around the cut piece **420**). Note that, basically, it is not necessary to enclose the arc-extinguishing material in the housing portion **310C**, but the arc-extinguishing material may be enclosed in the housing portion **310C** depending on the specification.

(135) Next, a state where the electric circuit breaker **600C** breaks an electric circuit in a case where an overcurrent belonging to a relatively low current range lower than a predetermined value flows through the electric circuit will be described with reference to FIG. 28. Note that FIG. 28(a) is a cross-sectional view illustrating a state where the breaker **970C** has moved from the state illustrated in FIG. 27(b), and FIG. 28(b) is a cross-sectional view illustrating a state where the moving body **500C** has moved from the state illustrated in FIG. 28(a).

(136) First, assuming that, when detecting an abnormal current in the electric circuit, the device **SC** determines that the abnormal current belongs to a relatively low current range lower than a predetermined value (for example, 1000 to 2000 A [amps]). Next, the device **SC** inputs the

abnormality signal X1C to the second power source 990C. As a result, the gunpowder in the second power source 990C explodes, and the air pressure due to the explosion is transmitted to the breaker 970C. The breaker 970C is then forcefully blown from the first end portion 320C toward the second end portion 330C by the air pressure, and instantaneously moves toward the second end portion 330 in the housing portion 380C. Then, as illustrated in FIG. 28(a), the cut piece 940C of the circuit 900C is strongly pushed downward by the projection 971C of the breaker 970C, and the cut piece 940C is cut in the vicinity of the coupling portion between the cut piece 940C and the base piece 930C and physically separated from the base piece 930C. As described above, since the circuit 900C is broken by the breaker 970C, the state where the fuse function portion 700C is electrically connected to the cut portion 400C via the circuit 900C is changed to a state where the fuse function portion 700C is not electrically connected to the cut portion 400C. As a result, the abnormal current I2C (see FIG. 27(a)) flowing through the base piece 430C does not flow through the circuit 900C in the fuse function portion 700C but flows only through the cut portion 400C. (137) Note that when the current belonging to the relatively low current range flows in the fuse function portion 700C through the circuit 900C, the fusion portion 740C of the fuse function portion 700C is not fused by the current belonging to the relatively low current range and thus the current cannot be broken, or it takes a relatively long time to break the current, and the overcurrent flowing through the electric circuit cannot be broken immediately.

(138) Thereafter, the device SC inputs the abnormality signal X2C to the first power source PC. As a result, the gunpowder in the first power source PC explodes, and the moving body 500C is forcefully blown from the first end portion 320C toward the second end portion 330C by the air pressure due to the explosion, and instantaneously moves toward the second end portion 330C in the housing portion 310C. Then, as illustrated in FIG. 28(b), the cut piece 420C is strongly pushed downward by the projection 530C of the moving body 500C, and the cut piece 420C is cut in the vicinity of the coupling portion between the cut piece 420C and the base piece 430C and physically separated from the base piece 430C. Therefore, the state where the base pieces 430C on both sides are energized is immediately broken, and the overcurrent I2C can be prevented from flowing through the electric circuit. Note that, since the abnormal current I2C belongs to the relatively low current range lower than the predetermined value, the arc discharge does not occur or the arc is immediately extinguished even if the distance between the cut piece 420C and the base piece 430C that are separated is short.

(139) Next, a state where the electric circuit breaker 600C breaks an electric circuit in a case where an overcurrent belonging to a relatively high current range higher than a predetermined value flows through the electric circuit will be described with reference to FIG. 29. Note that FIG. 29 is a cross-sectional view illustrating a state where the moving body 500C has moved from the state illustrated in FIG. 27(b).

(140) First, assuming that, when detecting an abnormal current in the electric circuit, the device SC determines that the abnormal current does not belong to a relatively low current range lower than a predetermined value but belongs to a relatively high current range higher than the predetermined value. Next, the device SC inputs the abnormality signal X2C only to the first power source PC without inputting the abnormality signal X1C to the second power source 990C.

(141) As a result, the gunpowder in the first power source PC explodes, and the moving body 500C instantaneously moves toward the second end portion 330C in the housing portion 310C. Then, as illustrated in FIG. 29, the moving body 500C moves toward the second end portion 330C, the cut piece 420C is strongly pushed downward by the projection 530C of the moving body 500C, and the cut piece 420C is cut in the vicinity of the coupling portion between the cut piece 420C and the base piece 430C and physically separated from the base piece 430C. That is, the state where the base pieces 430C on both sides of the cut portion 400C are energized via the cut piece 420C is broken, and an overcurrent can be prevented from flowing through the electric circuit.

(142) In addition, since the overcurrent belonging to the relatively high current range flows through

the base pieces **430C** on both sides connected to the electric circuit, an arc is possibly generated between the base piece **430C** and the cut piece **420C** immediately after cutting. However, as illustrated in FIGS. **27(a)** and **29**, since the cut piece **940C** of the circuit **900C** is not cut by the breaker **970C**, the fuse function portion **700C** is electrically connected to the cut portion **400C** via the circuit **900C**. Since the cut piece **420C** of the cut portion **400C** is cut while the fuse function portion **700C** and the cut portion **400C** remain electrically connected to each other, when the cut piece **420C** is cut, a current **I3C** (fault current) flowing through the electric circuit is induced in the fuse function portion **700C** via the circuit **900C** (see FIG. **27(a)**). Therefore, it is possible to prevent an arc from being generated between the base piece **430C** and the cut piece **420C**, which has been cut.

(143) The current **I3C** induced in the fuse function portion **700C** then causes the fusion portion **740C** of the fuse function portion **700C** to generate heat and fuse. Note that, when the cut piece **420C** is cut by the moving body **500C** to break the electric circuit, the current **I3C** is induced in the fuse function portion **700C** to flow through the electric circuit. Therefore, strictly speaking, the electric circuit is not completely broken. However, since the rating of the fusion portion **740C** of the fuse function portion **700C** is reduced, the fusion portion **740C** is immediately fused by the current **I3C**, and the electric circuit is immediately completely broken.

(144) Furthermore, after the fusion portion **740C** is fused, an arc is generated between the terminals **750C** of the fuse function portion **700C** by the voltage applied to the base pieces **430C** on both sides connected to the electric circuit, but the arc is quickly and effectively extinguished by the arc-extinguishing material **730C** in the fuse function portion **700C**.

(145) As described above, according to the electric circuit breaker **600C** of the present invention, a relatively high current (fault current) flowing through the electric circuit when the electric circuit is broken is induced in the fuse function portion **700C**, and the arc generated by the induced current can be effectively and quickly extinguished in the fuse function portion **700C**. As a result, it is possible to prevent the electric circuit breaker **600C** from being damaged by the arc generated between the base pieces **430C** in the housing **300C**, and to safely break the electric circuit.

(146) As described above, according to the electric circuit breaker **600C** of the present invention, in a case where the overcurrent belonging to the relatively low current range flows through the electric circuit, as illustrated in FIG. **28**, under a state where the cut portion **400C** and the fuse function portion **700C** are not connected, the cut piece **420C** between the base pieces **430C** on both sides of the cut portion **400C** is cut and the state where the base pieces **430C** on both sides are energized is immediately broken, so that the overcurrent is prevented from flowing through the electric circuit. Therefore, it is possible to solve the problem that, as in a conventional case, the current belonging to the relatively low current range cannot be broken because the fusion portion **740C** of the fuse function portion **700C** is not fused or the overcurrent flowing through the electric circuit cannot be broken immediately because it takes a relatively long time to break the current. On the other hand, in a case where the overcurrent belonging to the relatively high current range flows through the electric circuit, as illustrated in FIG. **29**, under a state where the cut portion **400C** and the fuse function portion **700C** are connected, the cut piece **420C** between the base pieces **430C** on both sides of the cut portion **400C** is cut and the state where the base pieces **430C** on both sides are energized is immediately and safely broken, so that the overcurrent is prevented from flowing through the electric circuit. As described above, the electric circuit breaker **600C** of the present invention quickly breaks current in a wide current range up to a relatively low current as well as a relatively high current.

Fifth Embodiment

(147) Next, an electric circuit breaker **600D** according to a fifth embodiment of the present invention will be described with reference to FIGS. **30** and **31**. In addition, since the configuration of the electric circuit breaker **600D** according to the fifth embodiment is basically the same as the configuration of the electric circuit breaker **600** according to the first embodiment except that the

regulating unit **800** is not provided, a circuit **900D** and a breaker **970D** are provided, and the configuration of a fuse function portion **700D**, the description of the same configuration will be omitted. Note that FIG. **30** is an overall perspective view illustrating the electric circuit breaker **600D** in an exploded manner, FIG. **31(a)** is a cross-sectional view taken along line N-N of FIG. **30**, and FIG. **31(b)** is a cross-sectional view taken along line M-M of FIG. **30**.

(148) As illustrated in FIGS. **30** and **31**, a lower housing **100D** is a substantially quadrangular prism formed of an insulator such as a synthetic resin, and includes a hollow lower housing portion **110D** therein. The lower housing portion **110D** is configured to house a moving body **500D**. In addition, the lower housing **100D** includes a hollow lower housing portion **160D** so as to be adjacent to the lower housing portion **110D**. The lower housing portion **160D** is configured to house the breaker **970D**.

(149) Furthermore, placement portions **113D** recessed based on the shape of a base piece **430D** are provided in a part of an upper surface **120D** of the lower housing **100D** so that the base piece **430D** of a cut portion **400D** can be placed. The placement portions **113D** are arranged on both sides of the lower housing portion **110D** so as to face each other, and support the cut portion **400D** extending linearly on both sides.

(150) In addition, the circuit **900D** is connected in parallel with the cut portion **400D**. The entire circuit **900D** is a metal conductor such as copper in order to be electrically connected to the cut portion **400D** via a fuse element **720D**. The circuit **900D** includes a base piece **930D** directly coupled to one base piece **430D** of the cut portion **400D** and another base piece **930D** directly coupled to the other base piece **430D** of the cut portion **400D**, and is coupled to the cut portion **400D** via the fuse element **720D**. More specifically, the linearly extending fuse element **720D** is inserted in a housing portion **972D** penetrating the breaker **970D** in a front-rear direction, and end portions **721D** on both sides of the fuse element **720D** projecting outward from the housing portion **972D** are individually coupled to the base pieces **930D**. The fuse element **720D** constitutes a part of the circuit **900D** and also constitutes a part of a fuse function portion to be described later.

Furthermore, placement portions **115D** recessed based on the shape of the base piece **930D** are provided in a part of the upper surface **120D** of the lower housing **100D** so that the base piece **930D** of the circuit **900D** can be placed. The placement portions **115D** are arranged on both sides of the lower housing portion **160D** so as to face each other, and support the circuit **900D** extending linearly on both sides.

(151) Furthermore, an upper housing **200D** is a substantially quadrangular prism formed of an insulator such as a synthetic resin, and forms a pair with the lower housing **100D** to constitute a housing **300D**. The upper housing includes a hollow upper housing portion **210D** therein, and the upper housing portion **210D** is configured to house the moving body **500D**. In addition, the upper housing **200D** includes a hollow upper housing portion **170D** so as to be adjacent to the upper housing portion **210D**. The upper housing portion **170D** is configured to house the breaker **970D**.

(152) Furthermore, insertion portions **213D** recessed based on the shape of the base piece **430D** are provided in a part of a lower surface **230D** of the upper housing **200D** so that the base piece **430D** of the cut portion **400D** can be inserted. The insertion portions **213D** are arranged on both sides of the upper housing portion **210D** so as to face each other, and are arranged at positions corresponding to the placement portions **113D** of the lower housing **100D**. Furthermore, insertion portions **215D** recessed based on the shape of the base piece **930D** are provided in a part of the lower surface **230D** of the upper housing **200D** so that the base piece **930D** of the circuit **900D** can be placed. The insertion portions **215D** are arranged on both sides of the upper housing portion **170D** so as to face each other, and support the circuit **900D** extending linearly on both sides.

(153) Moreover, a power source storage portion **221D** in which a first power source PD is housed is formed in a part of the side of an upper surface **220D** of the upper housing **200D**. The power source storage portion **221D** communicates with the upper end side of the upper housing portion **210D**. When it is detected that an abnormal current flows through the electric circuit, an

abnormality signal is input from an external device to the first power source PD. Then, for example, the gunpowder in the first power source PD is exploded, and the moving body **500D** is instantaneously pushed out and moved in a housing portion **310D** including the upper housing portion **210D** and the lower housing portion **110D** by the air pressure due to the explosion. Note that the housing portion **310D** extends from a first end portion **320D** of the housing **300D** to a second end portion **330D** opposite to the first end portion **320D**. Since the moving body **500D** is disposed on the side of the first end portion **320D**, the moving body can move toward the second end portion **330D** in the housing portion **310D** by the first power source PD provided on the side of the first end portion **320D**.

(154) Moreover, a power source storage portion **241D** in which a second power source **990D** is housed is formed in a part of the side of the upper surface **220D** of the upper housing **200D**. When it is detected that an abnormal current flows through the electric circuit, an abnormality signal is input from an external device to the second power source **990D**. Then, for example, the gunpowder in the second power source **990D** is exploded, and the breaker **970D** is instantaneously pushed out and moved in a housing portion **380D** including the upper housing portion **170D** and the lower housing portion **160D** by the air pressure due to the explosion. Note that the housing portion **380D** extends from the first end portion **320D** of the housing **300D** to the second end portion **330D** opposite to the first end portion **320D**. Since the breaker **970D** is disposed on the side of the first end portion **320D**, the breaker can move toward the second end portion **330D** in the housing portion **380D** by the second power source **990D** provided on the side of the first end portion **320D**.

(155) The electric circuit breaker **600D** also includes the fuse element **720D**. The periphery of the fuse element **720D** is filled with a granular arc-extinguishing material **730D**. In addition, one end portion **721D** of the fuse element **720D** is connected to the base piece **930D** directly coupled to the base piece **430D** of the cut portion **400D**, and the other end portion **721D** of the fuse element **720D** is connected to another base piece **930D** of the circuit **900D**. Therefore, the fuse element **720D** is electrically connected in parallel with the cut portion **400D** via the circuit **900D**. In addition, the fuse element **720D** includes a fusion portion **740D** between both ends, and the fusion portion **740D** is a portion in which the width of the fuse element **720D** is locally narrowed, and is configured to generate heat and fuse to break the current when the current to be broken by the electric circuit breaker flows. Note that the fuse element **720D** is housed in the housing portion **972D** of the breaker **970D**. The housing portion **972D** is filled with the arc-extinguishing material **730D** so as to surround the fuse element **720D**. The fuse function portion of the electric circuit breaker **600D** is different from the fuse function portion **700C** illustrated in FIG. 26, which is configured as a fuse in which the arc-extinguishing material **730C** and the fuse element **720C** are enclosed in the casing **710C**, in that the fuse function portion includes the fuse element **720D** including the fusion portion **740D** and the arc-extinguishing material **730D** filled in the housing portion **972D** of the breaker **970D**.

(156) Further, as illustrated in FIG. 31, the electric circuit breaker **600D** is attached in an electric circuit to be protected and used. Specifically, the base piece **430D** of the cut portion **400D** is connected to a part of the electric circuit, and the cut portion **400D** constitutes a part of the electric circuit. In addition, a projection **530D** of the moving body **500D** extends along the cut piece **420D** and is disposed away from the cut piece **420D**. In a normal state, since the base piece **430D** and the cut piece **420D** of the cut portion **400D** are not cut and are physically and electrically connected, a current **I1D** flows through the electric circuit via the base piece **430D** and the cut piece **420D** of the cut portion **400D**.

(157) In addition, in the normal state, the housing portion **972D** of the breaker **970D** is filled with the arc-extinguishing material **730D** so as to surround the fuse element **720D**, and the fuse element **720D** connects the two base pieces **930D**. That is, the circuit **900D** is neither disconnected nor broken by the breaker **970D**. Note that the resistance value of the fuse element **720D** is larger than the resistance value of the cut portion **400D**. Since the current **I1D** flowing through the cut portion

400D and a current **I1D'** flowing through the fuse element **720D** are proportional to the reciprocal of each resistance value, the magnitude of the current **I1D'** in the normal state is as small as about ten percent of the total current (current **I1D**+current **I1D'**).

(158) Furthermore, as illustrated in FIG. 31, a device **SD** that detects an abnormal current in the electric circuit is connected to the electric circuit to be protected. When detecting an abnormal current in the electric circuit by a built-in current sensor, an external current sensor connected to the electric circuit, or the like, the device **SD** determines whether or not the abnormal current belongs to a relatively low current range lower than a predetermined value (for example, 1000 to 2000 A [amps]). When determining that the abnormal current belongs to the relatively low current range lower than the predetermined value, the device **SD** inputs an abnormality signal **X1D** to the second power source **990D**. Thereafter, after a predetermined time has elapsed, the device **SD** inputs an abnormality signal **X2D** to the first power source **PD**. Note that, as described later, the predetermined time is a time until the breaker **970D** cuts the fuse element **720D** of the circuit **900D** by the second power source **990D**. On the other hand, when determining that the detected abnormal current does not belong to the relatively low current range lower than the predetermined value and belongs to a relatively high current range higher than the predetermined value, the device **SD** inputs the abnormality signal **X2D** only to the first power source **PD** without inputting the abnormality signal **X1D** to the second power source **990D**.

(159) Note that, as described later, in a case where a relatively high current flows, the electric circuit breaker **600D** can induce an arc generated when the electric circuit is broken in the fuse element **720D** to effectively and quickly extinguish the arc. Therefore, an arc-extinguishing material for extinguishing the arc is not enclosed in the housing portion **310D** (in particular, around the cut piece **420D**). Note that, basically, it is not necessary to enclose the arc-extinguishing material in the housing portion **310D**, but the arc-extinguishing material may be enclosed in the housing portion **310D** depending on the specification.

(160) Next, a state where the electric circuit breaker **600D** breaks an electric circuit in a case where an overcurrent belonging to a relatively low current range lower than a predetermined value flows through the electric circuit will be described with reference to FIG. 32. Note that FIG. 32(a) is a cross-sectional view illustrating a state where the breaker **970D** has moved from the state illustrated in FIG. 31(b), and FIG. 32(b) is a cross-sectional view illustrating a state where the moving body **500D** has moved from the state illustrated in FIG. 32(a).

(161) First, assuming that, when detecting an abnormal current in the electric circuit, the device **SD** determines that the abnormal current belongs to a relatively low current range lower than a predetermined value (for example, 1000 to 2000 A [amps]). Next, the device **SD** inputs the abnormality signal **X1D** to the second power source **990D**. As a result, the gunpowder in the second power source **990D** explodes, and the air pressure due to the explosion is transmitted to the breaker **970D**. The breaker **970D** is then forcefully blown from the first end portion **320D** toward the second end portion **330D** by the air pressure, and instantaneously moves toward the second end portion **330D** in the housing portion **380D**. Then, as illustrated in FIG. 32(a), the fuse element **720D** of the circuit **900D** is strongly pushed downward via the arc-extinguishing material **730D** by the breaker **970D**, and the fuse element **720D** is cut and physically separated from the base piece **930D**. As described above, since the circuit **900D** is broken by the breaker **970D**, it is changed to a state of not being electrically connected. As a result, an abnormal current **I2D** (see FIG. 31(a)) flowing through the base piece **430D** does not flow through the circuit **900D** but flows only through the cut portion **400D**.

(162) Note that when the current belonging to the relatively low current range flows in the fuse element **720D** through the circuit **900D**, the fuse element **720D** is not fused by the current belonging to the relatively low current range and thus the current cannot be broken, or it takes a relatively long time to break the current, and the overcurrent flowing through the electric circuit cannot be broken immediately.

(163) Thereafter, the device SD inputs the abnormality signal X2D to the first power source PD. As a result, the gunpowder in the first power source PD explodes, and the moving body 500D is forcefully blown from the first end portion 320D toward the second end portion 330D by the air pressure due to the explosion, and instantaneously moves toward the second end portion 330D in the housing portion 310D. Then, as illustrated in FIG. 32(b), the cut piece 420D is strongly pushed downward by the projection 530D of the moving body 500D, and the cut piece 420D is cut in the vicinity of the coupling portion between the cut piece 420D and the base piece 430D and physically separated from the base piece 430D. Therefore, the state where the base pieces 430D on both sides are energized is immediately broken, and the overcurrent I2D can be prevented from flowing through the electric circuit. Note that, since the abnormal current I2D belongs to the relatively low current range lower than the predetermined value, the arc discharge does not occur or the arc is immediately extinguished even if the distance between the cut piece 420D and the base piece 430D that are separated is short.

(164) Next, a state where the electric circuit breaker 600D breaks an electric circuit in a case where an overcurrent belonging to a relatively high current range higher than a predetermined value flows through the electric circuit will be described with reference to FIG. 33. Note that FIG. 33 is a cross-sectional view illustrating a state where the moving body 500D has moved from the state illustrated in FIG. 31(b).

(165) First, assuming that, when detecting an abnormal current in the electric circuit, the device SD determines that the abnormal current does not belong to a relatively low current range lower than a predetermined value but belongs to a relatively high current range higher than the predetermined value. Next, the device SD inputs the abnormality signal X2D only to the first power source PD without inputting the abnormality signal X1D to the second power source 990D.

(166) As a result, the gunpowder in the first power source PD explodes, and the moving body 500D instantaneously moves toward the second end portion 330D in the housing portion 310D. Then, as illustrated in FIG. 33, the moving body 500D moves toward the second end portion 330D, the cut piece 420D is strongly pushed downward by the projection 530D of the moving body 500D, and the cut piece 420D is cut in the vicinity of the coupling portion between the cut piece 420D and the base piece 430D and physically separated from the base piece 430D. That is, the state where the base pieces 430D on both sides of the cut portion 400D are energized via the cut piece 420D is broken, and an overcurrent can be prevented from flowing through the electric circuit.

(167) In addition, since the overcurrent belonging to the relatively high current range flows through the base pieces 430D on both sides connected to the electric circuit, an arc is possibly generated between the base piece 430D and the cut piece 420D immediately after cutting. However, as illustrated in FIGS. 31(a) and 33, since the fuse element 720D of the circuit 900D is not cut by the breaker 970D, the fuse element is electrically connected to the cut portion 400D. Since the cut piece 420D of the cut portion 400D is cut in this state, when the cut piece 420D is cut, a current I3D (fault current) flowing through the electric circuit is induced in the fuse element 720D via the circuit 900D (see FIG. 31(a)). Therefore, it is possible to prevent an arc from being generated between the base piece 430D and the cut piece 420D, which has been cut.

(168) The current I3D induced in the fuse element 720D then causes the fusion portion 740D of the fuse element 720D to generate heat and fuse. Note that, when the cut piece 420D is cut by the moving body 500D to break the electric circuit, the current I3D is induced in the fuse element 720D to flow through the electric circuit. Therefore, strictly speaking, the electric circuit is not completely broken. However, since the rating of the fusion portion 740D of the fuse element 720D is reduced, the fusion portion 740D is immediately fused by the current I3D, and the electric circuit is immediately completely broken.

(169) Furthermore, after the fusion portion 740D is fused, an arc is generated between the terminals 721D of the fuse element 720D by the voltage applied to the base pieces 430D on both sides connected to the electric circuit, but the arc is quickly and effectively extinguished by the arc-

extinguishing material **730D** in the housing portion **972D** of the breaker **970D**.

(170) As described above, according to the electric circuit breaker **600D** of the present invention, a relatively high current (fault current) flowing through the electric circuit when the electric circuit is broken is induced in the fuse element **720D** of the fuse function portion, and the arc generated by the induced current can be effectively and quickly extinguished by the arc-extinguishing material **730D**. As a result, it is possible to prevent the electric circuit breaker **600D** from being damaged by the arc generated between the base pieces **430D** in the housing **300D**, and to safely break the electric circuit.

(171) As described above, according to the electric circuit breaker **600D** of the present invention, in a case where the overcurrent belonging to the relatively low current range flows through the electric circuit, as illustrated in FIG. **32**, under a state where the cut portion **400D** and the fuse element **720D** are not connected, the cut piece **420D** between the base pieces **430D** on both sides of the cut portion **400D** is cut and the state where the base pieces **430D** on both sides are energized is immediately broken, so that the overcurrent is prevented from flowing through the electric circuit. Therefore, it is possible to solve the problem that, as in a conventional case, the current belonging to the relatively low current range cannot be broken because the fusion portion **740D** of the fuse element **720D** is not fused or the overcurrent flowing through the electric circuit cannot be broken immediately because it takes a relatively long time to break the current. On the other hand, in a case where the overcurrent belonging to the relatively high current range flows through the electric circuit, as illustrated in FIG. **33**, under a state where the cut portion **400D** and the fuse element **720D** are connected, the cut piece **420D** between the base pieces **430D** on both sides of the cut portion **400D** is cut and the state where the base pieces **430D** on both sides are energized is immediately and safely broken, so that the overcurrent is prevented from flowing through the electric circuit. As described above, the electric circuit breaker **600D** of the present invention quickly breaks current in a wide current range up to a relatively low current as well as a relatively high current.

(172) In addition, the electric circuit breaker of the present invention is not limited to the above embodiments, and various modifications and combinations are possible within the scope of the claims and the scope of the embodiments, and these modifications and combinations are also included in the scope of rights.

Claims

1. An electric circuit breaker that includes a housing, a cut portion that is disposed in the housing and constitutes a part of an electric circuit, a first power source that is disposed on a first end portion side of the housing, and a moving body that moves in the housing between the first end portion and a second end portion opposite to the first end portion, the electric circuit breaker comprising a fuse function portion that includes a fusion portion and an arc-extinguishing material, wherein the moving body is configured to cut a cut piece positioned between base pieces on two sides of the cut portion at a part of the moving body while moving from the first end portion toward the second end portion by the first power source, in a case where a current to be broken is low, the fuse function portion and the cut portion are not connected, the moving body is moved toward the second end portion by the first power source to cut the cut piece positioned between the base pieces on both sides of the cut portion to break a state where the base pieces on both sides of the cut portion are energized, and in a case where the current to be broken is high, the fuse function portion and the cut portion are put to the state of connection with each other, the moving body is moved toward the second end portion by the first power source to cut the cut piece positioned between the base pieces on both sides of the cut portion to break the state where the base pieces on both sides of the cut portion are energized; and further comprising paired electrodes individually connected to terminals on both sides of the fuse function portion, wherein in a case where a current to be broken

is low, the moving body moves toward the second end portion to cut a cut piece positioned between base pieces on both sides of the cut portion so as to break a state where the base pieces on both sides of the cut portion are energized, and a regulating unit operated by a second power source regulates movement of the moving body so as not to connect a part of the cut portion and the electrode in order to make the cut portion and the fuse function portion unconnected, in a case where the current to be broken is high, the moving body moves toward the second end portion, and in a state where the base pieces on both sides of the cut portion are energized via the cut piece, a part of the cut portion and the electrode come into contact with each other to connect the cut portion and the fuse, and thereafter, the state where the base pieces on both sides of the cut portion are energized via the cut piece is broken along with the movement of the moving body.

2. The electric circuit breaker according to claim 1, wherein the moving body includes the electrode, a state where base pieces on both sides of the cut portion are energized via the cut piece is a state where the base piece and the cut piece physically cut and separated from the base piece are energized by arc discharge, and the energized state is broken by an insulator being interposed between the base piece and the cut piece along with movement of the moving body.

3. The electric circuit breaker according to claim 1, wherein the housing includes the electrode, a state where base pieces on both sides of the cut portion are energized via the cut piece is a state where the base piece and the cut piece physically cut and separated from the base piece are energized by a conductor included in the moving body, and in the energized state, the base piece of the cut portion and the electrode are connected via the conductor of the moving body, and the cut portion and the fuse are connected.

4. An electric circuit breaker that includes a housing, a cut portion that is disposed in the housing and constitutes a part of an electric circuit, a first power source that is disposed on a first end portion side of the housing, and a moving body that moves in the housing between the first end portion and a second end portion opposite to the first end portion, the electric circuit breaker comprising a fuse function portion that includes a fusion portion and an arc-extinguishing material, wherein the moving body is configured to cut a cut piece positioned between base pieces on two sides of the cut portion at a part of the moving body while moving from the first end portion toward the second end portion by the first power source, in a case where a current to be broken is low, the fuse function portion and the cut portion are not connected, the moving body is moved toward the second end portion by the first power source to cut the cut piece positioned between the base pieces on both sides of the cut portion to break a state where the base pieces on both sides of the cut portion are energized, and in a case where the current to be broken is high, the fuse function portion and the cut portion are put to the state of connection with each other, the moving body is moved toward the second end portion by the first power source to cut the cut piece positioned between the base pieces on both sides of the cut portion to break the state where the base pieces on both sides of the cut portion are energized; and further comprising a circuit connected to the cut portion via the fuse function portion, wherein in a case where a current to be broken is low, the circuit is broken by a breaker moved by a second power source to be in a state where the fuse function portion and the cut portion are not connected, and thereafter, the moving body is moved toward the second end portion by a first power source to cut a cut piece positioned between base pieces on both sides of the cut portion so as to break a state where the base pieces on both sides of the cut portion are energized, and in a case where the current to be broken is high, in a state where the circuit is not broken and the fuse function portion and the cut portion remain connected to each other, the moving body is moved toward the second end portion by the first power source to cut the cut piece positioned between the base pieces on both sides of the cut portion so as to break the state where the base pieces on both sides of the cut portion are energized.

5. The electric circuit breaker according to claim 4, wherein a fuse element of the fuse function portion constitutes a part of the circuit, further, the fuse element is surrounded by an arc-

extinguishing material, and in a case where a current to be broken is low, the fuse element that is a part of the circuit is broken by a breaker moved by the second power source.
